Making a Difference?

The Role of Social Impact in College Major Choice

Ana Brás-Monteiro, Sara Rabino, Samantha Stelnicki

ZWEP - November 6, 2026

Constitution of the Italian Republic

"Italy is a Democratic Republic founded on labor."

"L'Italia è una Repubblica democratica, fondata sul lavoro."

Art. 1

Constitution of the Italian Republic

"Italy is a Democratic Republic founded on labor."

"L'Italia è una Repubblica democratica, fondata sul lavoro."

Art. 1

"Every citizen has the duty, in accordance with their capability and choice, to perform an activity or function that contributes to the material or spiritual progress of society."

"Ogni cittadino ha il dovere di svolgere, secondo le proprie possibilità e la propria scelta, un'attività o una funzione che concorra al progresso materiale o spirituale della società."

Art. 4

Constitution of the Italian Republic

What Jobs Contributes to Society?

What Jobs Contributes to Society?

"As a dietitian I help those with neurological disease improve their nutritional status and prevent malnutrition."

B.Sc. in **Nutrition**

What Jobs Contributes to Society?

"As a dietitian I help those with neurological disease improve their nutritional status and prevent malnutrition."

B.Sc. in **Nutrition**

"By preventing financial mistakes through software, detecting fraud, and ensuring audit compliance, the software I work with contributes to the financial security of millions of Americans."

B.Sc. in **Data Analytics**

What Jobs Contributes to Society?

"As a dietitian I help those with neurological disease improve their nutritional status and prevent malnutrition."

B.Sc. in **Nutrition**

"By preventing financial mistakes through software, detecting fraud, and ensuring audit compliance, the software I work with contributes to the financial security of millions of Americans."

B.Sc. in **Data Analytics**

"My job contributes to the public safety of those who take our medicines. ***I never expected when earning my chemistry degree that I would be in a role to protect the public in such a way.****"*

B.Sc. in **Chemistry**

What Jobs Contributes to Society?

"As a dietitian I help those with neurological disease improve their nutritional status and prevent malnutrition."

B.Sc. in **Nutrition**

"By preventing financial mistakes through software, detecting fraud, and ensuring audit compliance, the software I work with contributes to the financial security of millions of Americans."

B.Sc. in **Data Analytics**

"My job contributes to the public safety of those who take our medicines. ***I never expected when earning my chemistry degree that I would be in a role to protect the public in such a way.****"*

B.Sc. in **Chemistry**

We asked students: *name examples of jobs that contribute to society.*

What Jobs Contributes to Society?

"As a dietitian I help those with neurological disease improve their nutritional status and prevent malnutrition."

B.Sc. in **Nutrition**

"By preventing financial mistakes through software, detecting fraud, and ensuring audit compliance, the software I work with contributes to the financial security of millions of Americans."

B.Sc. in **Data Analytics**

"My job contributes to the public safety of those who take our medicines. ***I never expected when earning my chemistry degree that I would be in a role to protect the public in such a way.***"

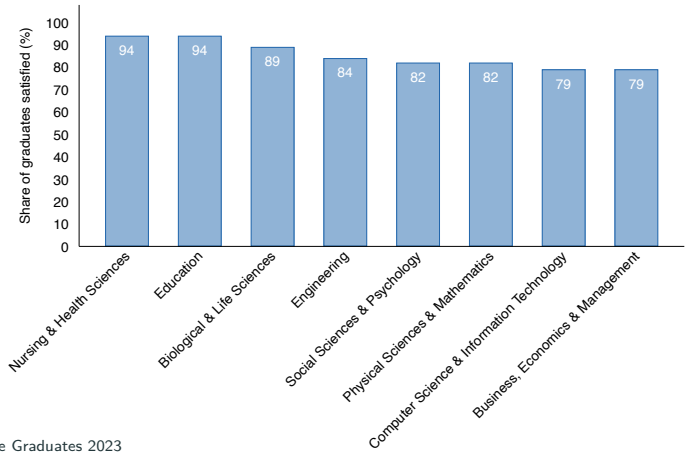
B.Sc. in **Chemistry**

We asked students: *name examples of jobs that contribute to society.*



Social Impact across Fields

How would you rate your satisfaction with the contribution to society of the job?



National Survey of College Graduates 2023

Stereotypes NSCG

This Paper

Experiment with undergraduate students at **Ohio State University**

This Paper

Experiment with undergraduate students at **Ohio State University**

1. Do students value the **social impact** of their future job?

This Paper

Experiment with undergraduate students at **Ohio State University**

1. Do students value the **social impact** of their future job?
    - Elicit preferences for future job attributes

This Paper

Experiment with undergraduate students at **Ohio State University**

1. Do students value the **social impact** of their future job?
   - Elicit preferences for future job attributes
2. How do students perceive the sense of contribution different fields provide? Do **stereotypes** shape these beliefs?

This Paper

Experiment with undergraduate students at **Ohio State University**

1. Do students value the **social impact** of their future job?
   - Elicit preferences for future job attributes
2. How do students perceive the sense of contribution different fields provide? Do **stereotypes** shape these beliefs?
   - Elicit beliefs about how pursuing different majors might lead to more socially impactful careers

This Paper

Experiment with undergraduate students at **Ohio State University**

1. Do students value the **social impact** of their future job?
   - Elicit preferences for future job attributes
2. How do students perceive the sense of contribution different fields provide? Do **stereotypes** shape these beliefs?
   - Elicit beliefs about how pursuing different majors might lead to more socially impactful careers
3. How do students' beliefs affect their **major choice**?
   - Model of college major choice
   - Exogenous shift in beliefs through information provision $\rightarrow$ measure effects on beliefs, recommendations and academic choices

Preview of Results

1. **Preferences:** contribution reported as important, **especially for women**

Preview of Results

1. **Preferences:** contribution reported as important, **especially for women**
2. **Beliefs:** stereotyped (right ranking, exaggerated gaps); Computer Science, Business & Econ, Physics & Mathematics the most underestimated

Preview of Results

1. **Preferences:** contribution reported as important, **especially for women**
2. **Beliefs:** stereotyped (right ranking, exaggerated gaps); Computer Science, Business & Econ, Physics & Mathematics the most underestimated
3. **Beliefs $\rightarrow$ choices:**
   - **Model:** one s.d. higher beliefs $\Rightarrow$ $+7$ percentage points (women), $+4$ (men)

Preview of Results

1. **Preferences:** contribution reported as important, **especially for women**
2. **Beliefs:** stereotyped (right ranking, exaggerated gaps); Computer Science, Business & Econ, Physics & Mathematics the most underestimated
3. **Beliefs $\rightarrow$ choices:**
   - **Model:** one s.d. higher beliefs $\Rightarrow$ $+7$ percentage points (women), $+4$ (men)
   - **Experiment:**
     - *Survey:* more diverse advice to other peers; for women, higher major intentions in their most underestimated fields
     - *University records:* for women, more credits in the three most underestimated fields, more Business minors, fewer Social Sciences majors

Related Literature

- College major choice (Patnaik et al., 2021; Wiswall and Zafar, 2015, 2018, 2021; Conlon, 2021; Conlon and Patel, 2026; Weinstein, 2022; Long et al., 2015)

  $\rightarrow$ ***Explore preferences and expectations for social impact***

Related Literature

- College major choice (Patnaik et al., 2021; Wiswall and Zafar, 2015, 2018, 2021; Conlon, 2021; Conlon and Patel, 2026; Weinstein, 2022; Long et al., 2015)

  → ***Explore preferences and expectations for social impact***

- Meaning at work (Brief and Nord, 1990; Cassar and Meier, 2018; Burbano et al., 2024b,a; Ariely et al., 2008; Ashraf et al., 2024)

  → ***Focus on students who yet have to decide their academic field***

Related Literature

- College major choice (Patnaik et al., 2021; Wiswall and Zafar, 2015, 2018, 2021; Conlon, 2021; Conlon and Patel, 2026; Weinstein, 2022; Long et al., 2015)

  → ***Explore preferences and expectations for social impact***

- Meaning at work (Brief and Nord, 1990; Cassar and Meier, 2018; Burbano et al., 2024b,a; Ariely et al., 2008; Ashraf et al., 2024)

  → ***Focus on students who yet have to decide their academic field***

- Stereotypes (Gennaioli and Shleifer, 2010; Bordalo et al., 2016, 2019; Conlon and Patel, 2026)

  → ***Apply stereotypes theory to college major choice and social impact***

Related Literature

- College major choice (Patnaik et al., 2021; Wiswall and Zafar, 2015, 2018, 2021; Conlon, 2021; Conlon and Patel, 2026; Weinstein, 2022; Long et al., 2015)

  → ***Explore preferences and expectations for social impact***

- Meaning at work (Brief and Nord, 1990; Cassar and Meier, 2018; Burbano et al., 2024b,a; Ariely et al., 2008; Ashraf et al., 2024)

  → ***Focus on students who yet have to decide their academic field***

- Stereotypes (Gennaioli and Shleifer, 2010; Bordalo et al., 2016, 2019; Conlon and Patel, 2026)

  → ***Apply stereotypes theory to college major choice and social impact***

- Under representation of women in STEM (Kahn and Ginther, 2018; Breda et al., 2023; Astorne-Figari and Speer, 2018)
  - Psychology (Su and Rounds, 2015; Diekman et al., 2011; Boucher et al., 2017)

  → ***Social impact beliefs as a driver of gender segregation***

Outline

1. Experimental Design
2. Preferences and Beliefs
3. Theoretical Framework
4. Treatment Effects
   - 4.1 Social Impact Beliefs
   - 4.2 Advice to Peers
   - 4.3 Academic Choices
5. Conclusions

Outline

1. **Experimental Design**

2. Preferences and Beliefs

3. Theoretical Framework

4. Treatment Effects

5. Conclusions

Empirical Setting

Ohio State University

- One of the largest universities in the U.S. by enrollment
- Students get to experiment two years before officially declaring a major
- Flexibility in changing major also after the second year

Experimental Design

The main survey is divided into five parts:

Experimental Design

The main survey is divided into five parts:

1. Measure **initial major intentions**:
    - Likelihood of graduating in a major in each field Question

Experimental Design

The main survey is divided into five parts:

1. Measure initial major intentions
2. Measure **preferences** for different job attributes, including contribution to society

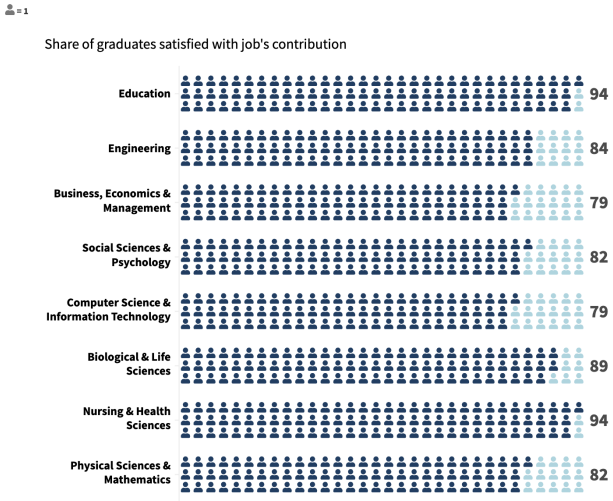
Experimental Design

The main survey is divided into five parts:

1. Measure initial major intentions
2. Measure preferences for different job attributes, including contribution to society
3. Measure **beliefs** about the social impact of different majors:
   - Career-related factors influencing major choice - **top of mind beliefs** Question
   - Likelihood of being satisfied with job's contribution to society if they graduate in each major - **self beliefs** Question
   - Share of previous graduates in each field who are satisfied with their current job's contribution to society - **population beliefs** Question

4. **Information intervention** on social impact for treated students only:
    - **Population statistics** for each major: share of previous graduates that are satisfied with their job's contribution from the 2023 National Survey of College Graduates (NSCG)

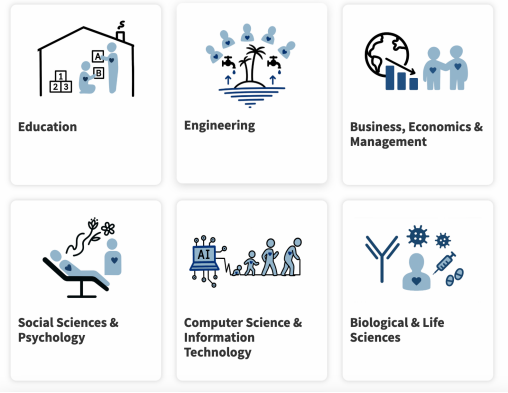
Treatment - Statistics

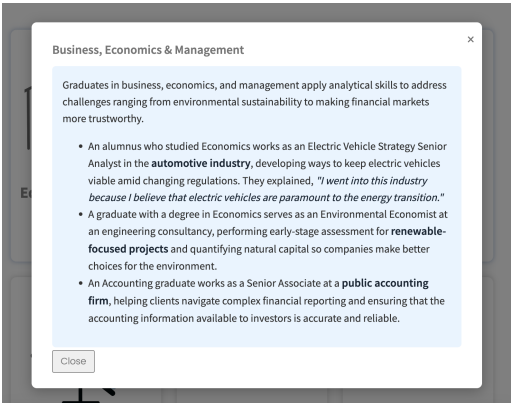


Experimental Design

4. **Information intervention** on social impact for treated students only:
   - **Population statistics** for each major: share of previous graduates that are satisfied with their job's contribution from the 2023 National Survey of College Graduates (NSCG)
   - **Stories** for each major: reflections from OSU alumni on the social impact of their job Alumni survey Validation survey

Treatment - Stories





Open interactive | Other stories

Experimental Design

4. **Information intervention** on social impact for treated students only:
    - **Population statistics** for each major: share of previous graduates that are satisfied with their job's contribution from the 2023 National Survey of College Graduates (NSCG)
    - **Stories** for each major: reflections from OSU alumni on the social impact of their current job
    - **Active control**: neutral OECD statistics unrelated to social impact

Experimental Design

4. Information intervention on social impact for treated students only
5. **Outcomes** of interest:
   - **Self beliefs**
   - **Major recommendation to peers** Question
   - **Major intentions**

Experimental Design

4. Information intervention on social impact for treated students only
5. **Outcomes** of interest:
   - **Self beliefs**
   - **Major recommendation to peers** Question
   - **Major intentions**

Access to **administrative data** from OSU Registrar's Office:

- Enrolled classes in Fall 2025, Spring 2026 and Fall 2026
- Major and Minor plans in Fall 2025, Spring 2026 and Fall 2026

Outline

1. Experimental Design

2. Preferences and Beliefs

3. Theoretical Framework

4. Treatment Effects

5. Conclusions

Our Sample

Undergraduate students at Ohio State: 657 valid responses

	N	%
Class year		
Freshman	314	47.8
Sophomore	304	46.3
Junior	39	5.9
Gender		
Female	420	63.9
Male	220	33.5
Other	12	1.8
Prefer not to say	5	0.8

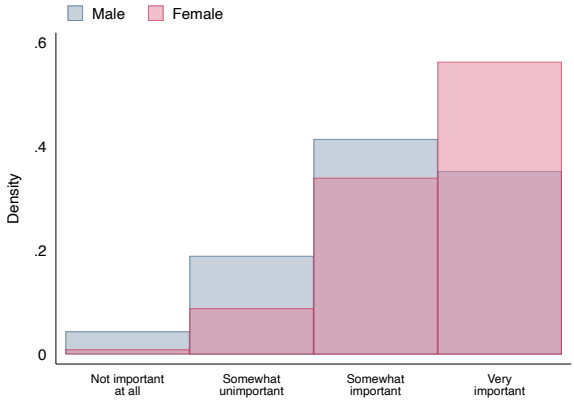
	N	%
Ethnicity		
White	390	59.4
Asian	122	18.6
Two or more	77	11.7
Black or African Am.	35	5.3
Hispanic	13	2.0
Prefer not to say	11	1.7
Other	8	1.2
Amer. Indian/Alaska	1	0.2

Unsure about major (>1 field): **71.4%** (median 85% on leading major)

Median completion time: ≈33 min

Social Impact Matters, But More So for Women

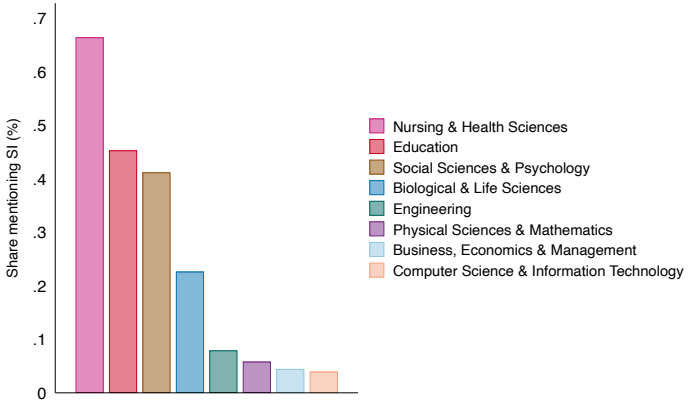
When thinking about a job, how important is each of the following factors to you?
"Contribution to Society"



Gender WTP

Beliefs: Top of Mind

For each field, think about the types of jobs you could get after that major. List a few aspects of those jobs that you would be satisfied with.



Beliefs: Self vs Population Questions

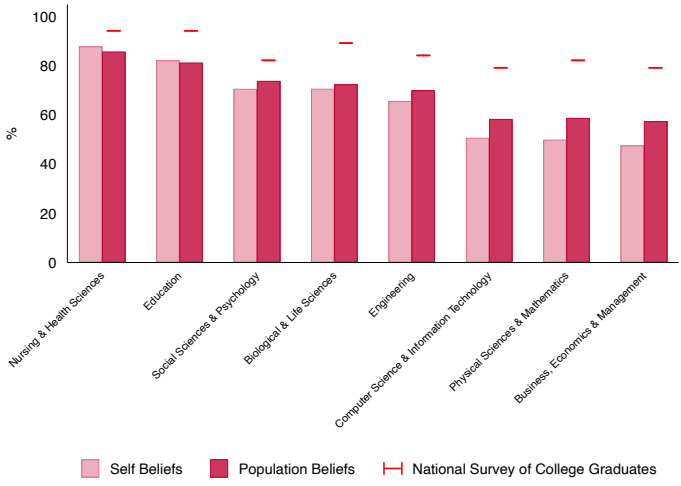
Self Beliefs Question

*"Suppose **you** received a Bachelor's degree in each of the following majors and were working full-time at age 30. What do you think is the percent chance that **you** would feel satisfied with **your** job's contribution to society?"*

Population Beliefs Question

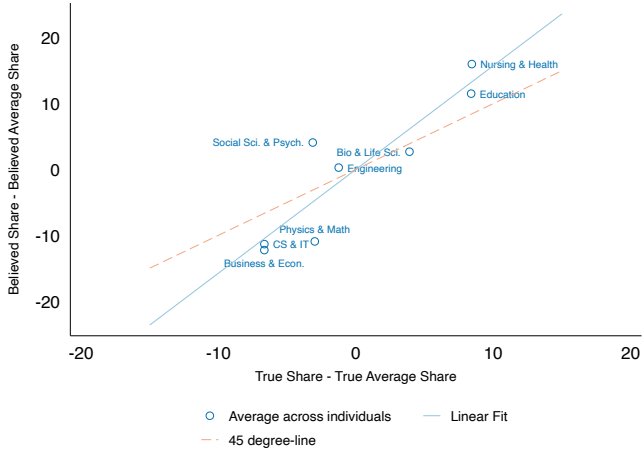
*"**Among all college graduates** aged 30 to 35 who work full-time and hold a Bachelor's degree in each of the following major groups, what percentage do you think feel satisfied with **their** job's contribution to society?"*

Own Expectations Mirror Beliefs about Graduates



Correlation | Gap and preferences | Gender

Differences between Fields Are Exaggerated



Regression Salary

Outline

1. Experimental Design

2. Preferences and Beliefs

3. Theoretical Framework

4. Treatment Effects

5. Conclusions

Basic Theoretical Framework

Student i's **expected utility** of student from choosing $m \in \{1, ..., M\}$ is:

$$EU^i[m] = \alpha_1 E^i[w_m] + \theta^i \tilde{\pi}^i_m + \alpha_2 E^i[GPA_m] + \mu^i_m + \epsilon^i_m$$

- wage w_m
- perceived probability of satisfaction with contribution $\tilde{\pi}^i_m$
- subjective ability $E^i[GPA_m]$
- non-ability benefits $\mu^i_m \sim \mathcal{N}(\mu_m, \sigma^2)$
- ϵ^i type 1 extreme value random variable

Basic Theoretical Framework

Student i's **expected utility** of student from choosing $m \in \{1, ..., M\}$ is:

$$EU^i[m] = \alpha_1 E^i[w_m] + \theta^i \tilde{\pi}^i_m + \alpha_2 E^i[GPA_m] + \mu^i_m + \epsilon^i_m$$

- wage w_m
- **perceived probability of satisfaction with contribution $\tilde{\pi}^i_m$**
- subjective ability $E^i[GPA_m]$
- non-ability benefits $\mu^i_m \sim \mathcal{N}(\mu_m, \sigma^2)$
- ϵ^i type 1 extreme value random variable

Basic Theoretical Framework

Social impact affects expected utility through two channels:

1. **Preferences θ^i**
2. **Beliefs $\tilde{\pi}^i_m$**

The probability of choosing major m instead of m':

$$\log \frac{q^i_m}{q^i_{m'}} = \alpha_1 \left(E^i[w_m] - E^i[w_{m'}] \right) + \boldsymbol{\theta^i} \left(\boldsymbol{\tilde{\pi}^i_m} - \boldsymbol{\tilde{\pi}^i_{m'}} \right)$$
$$+\alpha_2 \left(E^i\left[GPA_m\right] - E^i\left[GPA_{m'}\right] \right) + \mu^i_m - \mu^i_{m'}.$$

We estimate with Maximum Likelihood the parameters $\gamma = (\theta, \alpha_1, \alpha_2, \mu_m, \sigma)$ More

Predictions

H1. Information on the true share of graduates satisfied with their contribution shifts students' own self-beliefs in the direction of their population misperception.

Predictions

H1. Information on the true share of graduates satisfied with their contribution shifts students' own self-beliefs in the direction of their population misperception.

H2. A shift in beliefs moves students toward the most misperceived majors: the *rank* of revisions in choice follows the *rank* of revisions in beliefs.

Beliefs Matter for Choice

Estimating the model on students' prior major intentions and beliefs recovers how much they weight each job attribute:

Gender

Beliefs Matter for Choice

Estimating the model on students' prior major intentions and beliefs recovers how much they weight each job attribute:

- Salary, social-impact beliefs, and GPA all significantly predict choice

	Winsor. prob. [0.5, 99.5]
α_1: Salary ($10k)	0.038***
	(0.013)
$\bar{\theta}$: Impact belief	2.939***
	(0.219)
Women	3.411***
	(0.254)
Men	2.045***
	(0.285)
α_2: E[GPA]	1.533***
	(0.074)
σ^2: Var(μ_m^i)	2.535
	(0.049)
WTP: 1 pp belief ($10k)	0.765***
	(0.269)
Median $\hat{\theta}_i$	2.829
Students	605

Gender

Beliefs Matter for Choice

Estimating the model on students' prior major intentions and beliefs recovers how much they weight each job attribute:

- Salary, social-impact beliefs, and GPA all significantly predict choice
- The impact weight is ∼65% larger for women ($\bar{\theta} \approx 3.4$) than men ($\bar{\theta} \approx 2.0$)
- WTP stable across winsorization bounds

	Winsor. prob. [0.5, 99.5]
α_1: Salary ($10k)	0.038***
	(0.013)
$\bar{\theta}$: Impact belief	2.939***
	(0.219)
Women	3.411***
	(0.254)
Men	2.045***
	(0.285)
α_2: E[GPA]	1.533***
	(0.074)
σ^2: Var(μ_m^i)	2.535
	(0.049)
WTP: 1 pp belief ($10k)	0.765***
	(0.269)
Median $\hat{\theta}_i$	2.829
Students	605

Gender

Beliefs Matter for Choice

Estimating the model on students' prior major intentions and beliefs recovers how much they weight each job attribute:

- Salary, social-impact beliefs, and GPA all significantly predict choice
- The impact weight is ∼65% larger for women ($\bar{\theta} \approx 3.4$) than men ($\bar{\theta} \approx 2.0$)
- WTP stable across winsorization bounds
- **Implied effect: a 1 s.d. rise in impact beliefs raises own-major probability by ≈7 p.p. (women) vs. ≈4 p.p. (men)**

	Winsor. prob. [0.5, 99.5]
α_1: Salary ($10k)	0.038***
	(0.013)
$\bar{\theta}$: Impact belief	2.939***
	(0.219)
Women	3.411***
	(0.254)
Men	2.045***
	(0.285)
α_2: E[GPA]	1.533***
	(0.074)
σ^2: Var(μ_m^i)	2.535
	(0.049)
WTP: 1 pp belief ($10k)	0.765***
	(0.269)
Median $\hat{\theta}_i$	2.829
Students	605

Gender

Predictions

H2. A shift in beliefs moves students toward the most misperceived majors: the *rank* of revisions in choice follows the *rank* of revisions in beliefs.

Predictions

H2. A shift in beliefs moves students toward the most misperceived majors: the *rank* of revisions in choice follows the *rank* of revisions in beliefs.

Imposing self-beliefs equal to graduates' levels reallocates choices (pp, women):

Away ↓		**Toward ↑**	
Social Sciences	−1.9	Business & Econ.	+2.3
Health	−1.7	Physics & Maths	+2
		Computer Science	+1.4

Counterfactual analysis

Preference counterfactual

Outline

1. Experimental Design

2. Preferences and Beliefs

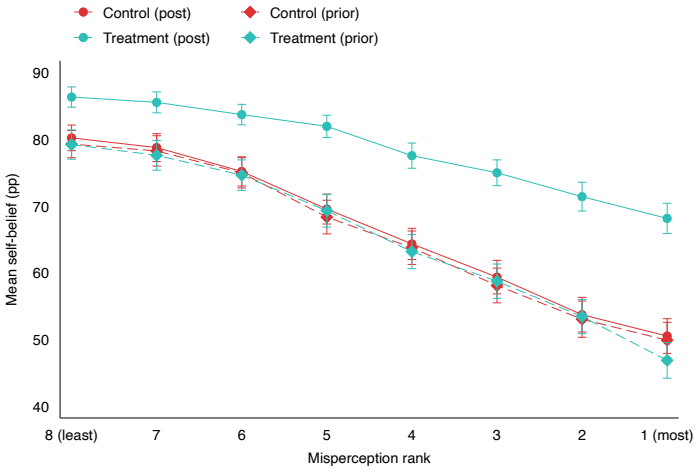
3. Theoretical Framework

4. Treatment Effects

- 4.1 Social Impact Beliefs
- 4.2 Advice to Peers
- 4.3 Academic Choices

5. Conclusions

Information Shrinks Perceived Differences across Fields

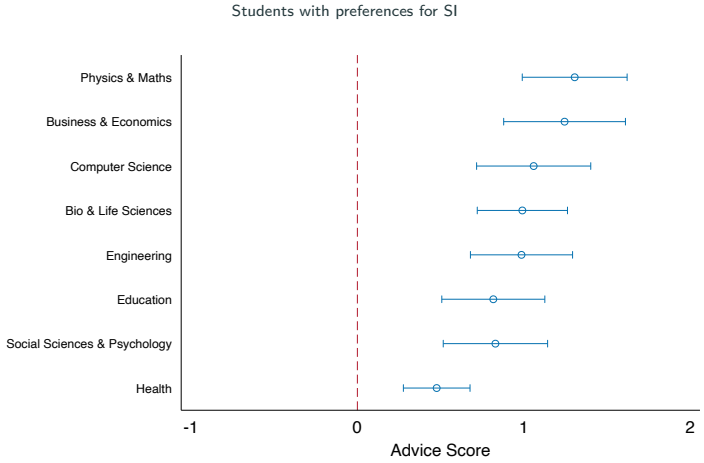


Regression SDB Self vs Pop

Treated Students Advise Peers to Pursue More Diverse Majors for Impact

Students with preferences for SI

Treated Students Advise Peers to Pursue More Diverse Majors for Impact



Question | Gender | Misperception

Do Students Move toward Fields They Underestimate?

Do Students Move toward Fields They Underestimate?

Pooling **all majors**:

$$\begin{aligned}\text{PostP}_{i,m} = \alpha + \beta_0\,\text{PreP}_{i,m} + \beta_1\,\text{Treat}_i + \beta_2\,\text{DemeanedUndergap}_{i,m} \\ + \beta_3\,\text{Treat}_i \cdot \text{DemeanedUndergap}_{i,m} + \beta_4\,\text{Credits}_{i,m} + \beta_5 X_i + \epsilon_{i,m}\end{aligned}$$

- $\text{PostP}_{i,m}$ and $\text{PreP}_{i,m}$: posterior and prior probabilities of individual i to pursue m

Do Students Move toward Fields They Underestimate?

Pooling **all majors**:

$$\begin{aligned}\text{PostP}_{i,m} = \alpha + \beta_0\,\text{PreP}_{i,m} + \beta_1\,\text{Treat}_i + \beta_2\,\text{DemeanedUndergap}_{i,m} \\ + \beta_3\,\text{Treat}_i \cdot \text{DemeanedUndergap}_{i,m} + \beta_4\,\text{Credits}_{i,m} + \beta_5 X_i + \epsilon_{i,m}\end{aligned}$$

- $\text{PostP}_{i,m}$ and $\text{PreP}_{i,m}$: posterior and prior probabilities of individual i to pursue m
- $\text{DemeanedUndergap}_{i,m}$: underestimation of individual i of the share of satisfied graduates in m, demeaned by the average underestimation across all majors

Do Students Move toward Fields They Underestimate?

Pooling **all majors**:

$$\begin{aligned}\text{PostP}_{i,m} = \alpha + \beta_0\,\text{PreP}_{i,m} + \beta_1\,\text{Treat}_i + \beta_2\,\text{DemeanedUndergap}_{i,m} \\ + \beta_3\,\text{Treat}_i \cdot \text{DemeanedUndergap}_{i,m} + \beta_4\,\text{Credits}_{i,m} + \beta_5 X_i + \epsilon_{i,m}\end{aligned}$$

- $\text{PostP}_{i,m}$ and $\text{PreP}_{i,m}$: posterior and prior probabilities of individual i to pursue m
- $\text{DemeanedUndergap}_{i,m}$: underestimation of individual i of the share of satisfied graduates in m, demeaned by the average underestimation across all majors
- Controls: credits in m in the previous semesters, age, year, gender and campus FEs

Do Students Move toward Fields They Underestimate?

Pooling **all majors**:

$$\begin{aligned}\text{PostP}_{i,m} = \alpha + \beta_0\,\text{PreP}_{i,m} + \beta_1\,\text{Treat}_i + \beta_2\,\text{DemeanedUndergap}_{i,m} \\ + \beta_3\,\text{Treat}_i \cdot \text{DemeanedUndergap}_{i,m} + \beta_4\,\text{Credits}_{i,m} + \beta_5 X_i + \epsilon_{i,m}\end{aligned}$$

- $\text{PostP}_{i,m}$ and $\text{PreP}_{i,m}$: posterior and prior probabilities of individual i to pursue m
- $\text{DemeanedUndergap}_{i,m}$: underestimation of individual i of the share of satisfied graduates in m, demeaned by the average underestimation across all majors
- Controls: credits in m in the previous semesters, age, year, gender and campus FEs

Hypothesis: $\beta_3 > 0$

Treated Women Are More Likely to Choose More Misperceived Fields

Students with preferences for SI

	Final P(M)
	(1)
	All
Initial P(M)	0.949***
	(0.012)
Treatment	0.098
	(0.115)
Demeaned Undergap	-0.066***
	(0.020)
Demeaned Undergap*Treatment	0.042**
	(0.021)
Prior credits in cat. (AU25+SP26)	0.124**
	(0.050)
Constant	0.109
	(0.764)
FEs	Yes
Observations	4496
Adjusted R^2	0.937

Control Other outcomes GPA Misperception Rank New Interest Self vs Pop

Treated Women Are More Likely to Choose More Misperceived Fields

Students with preferences for SI

	Final P(M)
	(1)
	All
Initial P(M)	0.949***
	(0.012)
Treatment	0.098
	(0.115)
Demeaned Undergap	-0.066***
	(0.020)
Demeaned Undergap*Treatment	0.042**
	(0.021)
Prior credits in cat. (AU25+SP26)	0.124**
	(0.050)
Constant	0.109
	(0.764)
FEs	Yes
Observations	4496
Adjusted R^2	0.937

Control Other outcomes GPA Misperception Rank New Interest Self vs Pop

Treated Women Are More Likely to Choose More Misperceived Fields

Students with preferences for SI

	Final P(M)	
	(1)	(2)
	All	Women
Initial P(M)	0.949***	0.955***
	(0.012)	(0.013)
Treatment	0.098	0.217
	(0.115)	(0.151)
Demeaned Undergap	-0.066***	-0.074***
	(0.020)	(0.025)
Demeaned Undergap*Treatment	0.042**	0.060**
	(0.021)	(0.024)
Prior credits in cat. (AU25+SP26)	0.124**	0.079
	(0.050)	(0.057)
Constant	0.109	-0.562
	(0.764)	(0.886)
FEs	Yes	Yes
Observations	4496	3104
Adjusted R^2	0.937	0.944

Control | Other outcomes | GPA | Misperception Rank | New Interest | Self vs Pop

Treated Women Are More Likely to Choose More Misperceived Fields

Students with preferences for SI

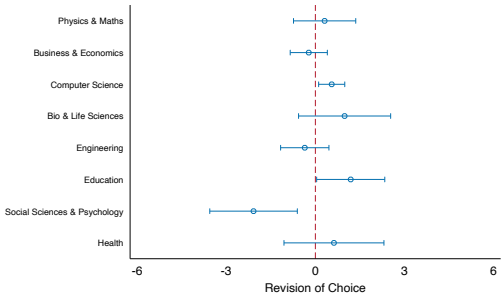
	Final P(M)		
	(1)	(2)	(3)
	All	Women	Men
Initial P(M)	0.949***	0.955***	0.936***
	(0.012)	(0.013)	(0.026)
Treatment	0.098	0.217	-0.127
	(0.115)	(0.151)	(0.155)
Demeaned Undergap	-0.066***	-0.074***	-0.045
	(0.020)	(0.025)	(0.035)
Demeaned Undergap*Treatment	0.042**	0.060**	0.003
	(0.021)	(0.024)	(0.041)
Prior credits in cat. (AU25+SP26)	0.124**	0.079	0.219**
	(0.050)	(0.057)	(0.101)
Constant	0.109	-0.562	3.077
	(0.764)	(0.886)	(2.324)
FEs	Yes	Yes	Yes
Observations	4496	3104	1392
Adjusted R^2	0.937	0.944	0.922

Control | Other outcomes | GPA | Misperception Rank | New Interest | Self vs Pop

On Aggregate Intentions Maps into Real Choices

Survey

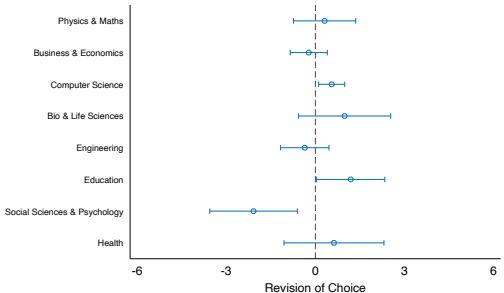
Treatment effect on reported intentions $P(\text{major})$



On Aggregate Intentions Maps into Real Choices

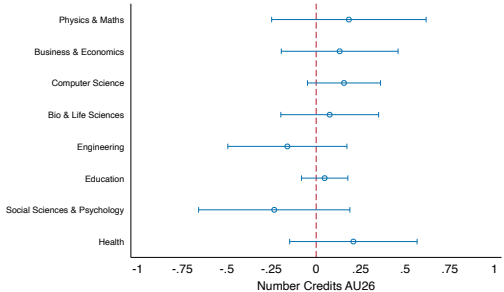
Survey

Treatment effect on reported intentions $P(\text{major})$



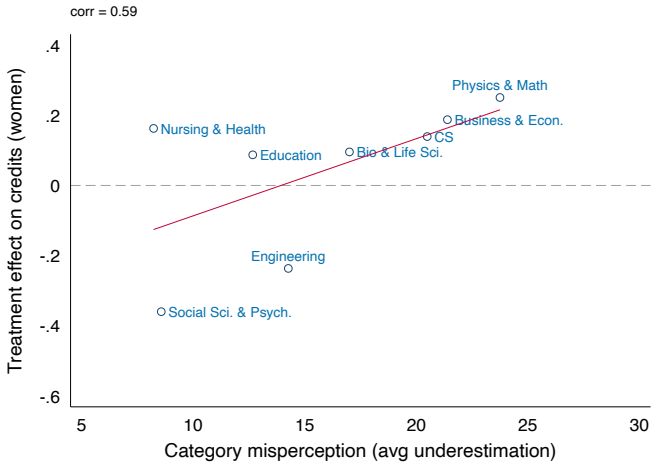
Administrative data

Treatment effect on # AU26 credits, by field



Gender | First snapshot

Treated Women Enroll in the Most Systematically Underestimated Fields



Top 3 Men Other admin results

Outline

1. Experimental Design

2. Preferences and Beliefs

3. Theoretical Framework

4. Treatment Effects

5. Conclusions

Summary

1 Preferences

- Most students value the social impact of their future jobs, women more so

Summary

1 Preferences

- Most students value the social impact of their future jobs, women more so

2 Beliefs

- Students **underestimate** graduates' satisfaction with their contribution
- Stronger in some majors $\rightarrow$ **stereotypes**

Summary

1 Preferences

- Most students value the social impact of their future jobs, women more so

2 Beliefs

- Students **underestimate** graduates' satisfaction with their contribution
- Stronger in some majors $\rightarrow$ **stereotypes**

3 Information

- Shifts beliefs and recommendations to a like-minded peer
- Shifts choice **only for women** who care about impact:
  - $\uparrow$ majors perceived as low-contribution
  - $\downarrow$ majors perceived as high-contribution

Thank you!



https://sararabino.github.io/

sara.rabino@econ.uzh.ch

What Contribution Means (to Students)

In your opinion, what does it mean for a job to contribute to society?

"to help others"
"a positive impact on people's lives"
"it makes people's lives better or easier"
"provides a valuable service"
"give back to the community"
"beyond just making money"
"make the world a better place"

Representativeness

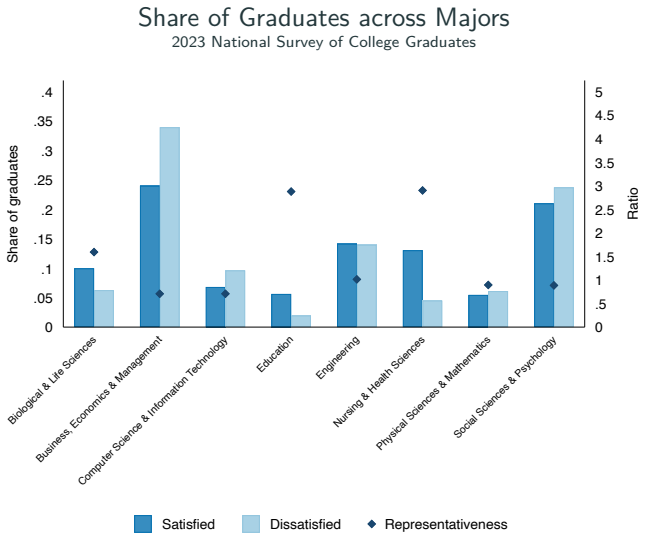
- Overall, 55% mentioned health-related jobs, 35% mentioned teaching, while 4% mentioned computing jobs and less than 1% mathematicians or economists
- Nursing & Health Sciences and Education are the most representative types for the group of graduates that are satisfied with their contribution to society
- The **representativeness** of major m for group G (satisfied with job's contribution to society) is defined as the likelihood ratio:

$$\frac{P(m|G)}{P(m|-G)}$$

where $P(m|G)$ is the probability that a college graduate that is satisfied with their job's contribution to society graduated in major m

Back

What Comes to Mind



Representativeness of Majors across Job Attributes

	Contrib.	Salary	Advanc.	Challenge	Benefits	Indep.	Location	Respons.	Security
Bio & Life Sci.	1.59	0.70	1.04	1.12	1.04	1.52	0.88	1.04	1.30
Business & Econ.	0.71	1.15	0.80	0.77	0.87	1.09	0.98	0.87	1.12
Computer Science	0.71	1.37	1.27	1.28	1.13	1.23	1.16	1.04	0.72
Education	**2.88**	0.46	2.27	2.10	0.50	1.45	2.15	1.19	0.95
Engineering	1.01	1.55	1.22	1.20	1.36	1.15	0.69	1.02	0.91
Health	**2.90**	0.89	1.01	1.62	0.96	1.27	1.22	1.24	1.47
Physics & Maths	0.89	1.06	1.09	0.87	1.42	1.30	1.17	1.23	0.76
Social Sci. & Psych.	0.89	0.93	0.90	0.82	1.09	0.59	1.06	0.95	0.88

Only for **contribution** do two majors (Education, Health) reach a representativeness of ≈ 3; for other attributes, gaps are much smaller.

Back

Overall Satisfaction

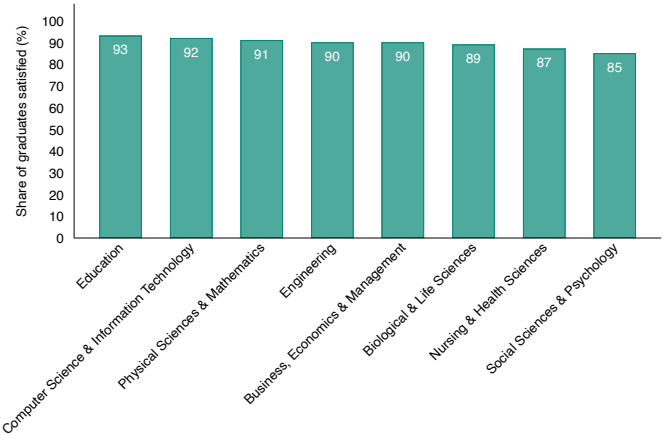
Table 1: Correlation Overall Satisfaction and Job Features

	(1) Overall Satisfaction	(2) Overall Satisfaction
Satisfaction Advancement	0.103**	0.101**
	(0.047)	(0.045)
Satisfaction Benefits	−0.009	−0.004
	(0.061)	(0.056)
Satisfaction Challenge	0.092**	0.089**
	(0.036)	(0.035)
Satisfaction Independence	0.161***	0.155***
	(0.026)	(0.026)
Satisfaction Location	0.066***	0.067***
	(0.017)	(0.017)
Satisfaction Responsibility	0.122***	0.125***
	(0.046)	(0.045)
Satisfaction Salary	0.280***	0.274***
	(0.045)	(0.040)
Satisfaction Security	0.096***	0.094***
	(0.027)	(0.026)
Satisfaction Contribution	0.154***	0.166***
	(0.024)	(0.027)
Major FEs	*No*	*Yes*
Observations	11699	11699
Adjusted R^2	0.541	0.547

Back

Overall Satisfaction across Fields

How would you rate your overall satisfaction with this job?



National Survey of College Graduates 2023

Back

Major Intentions

What do you believe is the **percent chance** (or chances out of 100) that you would either **graduate** with a major in the following major categories or that you would never graduate/drop-out?

Even if you have declared or plan to declare a major, you might still be uncertain about which one you will ultimately graduate with. This question helps us understand the majors you are currently considering in your choice.

Note: Your answers to the major categories should sum to 100.

▶ *Click for more info on major groups*

Education	0
Computer Science & Information Technology	0
Biological & Life Sciences	0
Nursing & Health Sciences	0
Social Sciences & Psychology	0
Physical Sciences & Mathematics	0
Engineering	0
Business, Economics & Management	0
Other	0
Never Graduate/Drop Out	0
Total	0

Back

OSU Alumni Survey

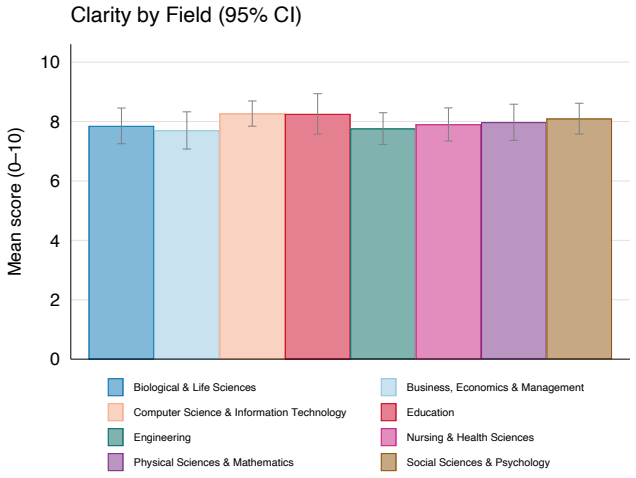
In 1-2 sentences, please describe what you do in your current role.

In your own words, how does your work **contribute to society** or help improve people's lives?

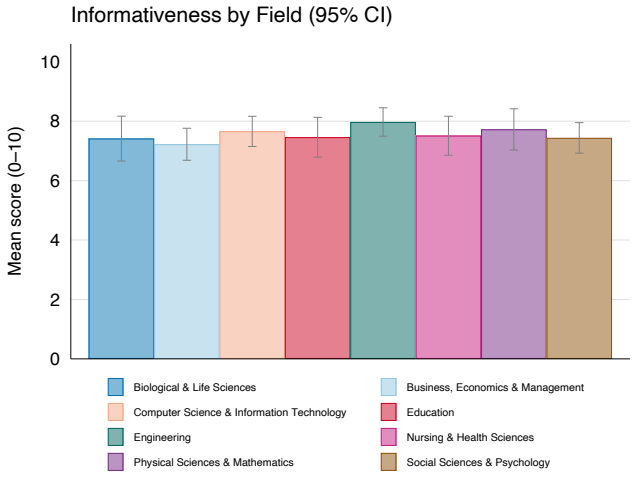
Can you describe a **moment**, **project**, or **example** that illustrates the positive impact of your work?

Back

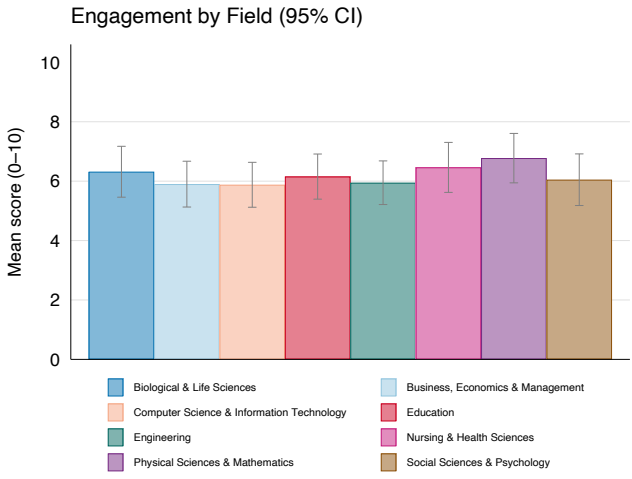
Prolific Validation Survey: Clarity



Prolific Validation Survey: Informativeness



Prolific Validation Survey: Engagement



Back

Beliefs - Top



Suppose you received a Bachelor's degree in each of the following major groups and were working full-time at age 30. **For each field**, think about the types of jobs you could get after that major. **List a few aspects of those jobs that you would be satisfied with.**

Keep your answers short — just short phrases or keywords, not full sentences.

▶ *Click for more info on major groups*

back (design) back (results)

Beliefs - Self



Suppose you received a Bachelor's degree in each of the following major groups and were working full-time at age 30. What do you think is the **percent chance** that you would feel **satisfied** with your **job's contribution to society**?

Using a scale from 0 to 100, enter the chance you would feel satisfied with your **job's contribution to society**.

Example:
Imagine you have graduated in the field of Law and are working full-time at age 30. Even if this is a not a major you're interested in, try to picture yourself in that situation. You may not be sure how you would feel about your job's contribution to society — it might depend on the role or organisation.
You might think:

- There's a **60% chance** you would feel **satisfied** with your job's **contribution to society**
- There's a **40% chance** you would feel **dissatisfied**

In this case, you would enter **60%** for Law.
This helps us understand how likely you think each outcome is — not just how you feel right now, but how certain or uncertain you are about how you might feel in the future.

back (design) back (results)

Beliefs - Population



Among all **college graduates aged 30–35** who work full-time and hold a Bachelor's degree in each of the following major groups, what **percentage** do you think feel **satisfied** with their **job's contribution to society**?

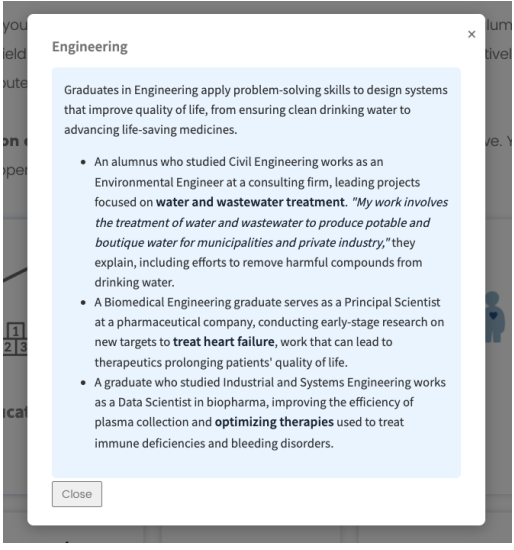
Using a scale from 0 to 100, enter the share of graduates who feel **satisfied** with their **job's contribution to society**.

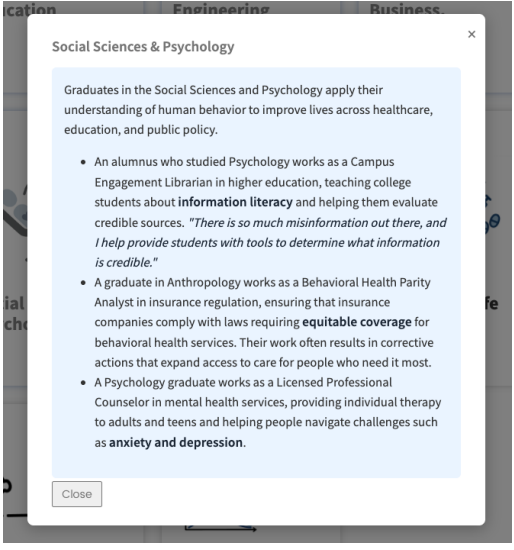
Example:
Suppose you think that **55%** of Law graduates feel **satisfied** with their **job's contribution to society**, while **45%** feel **dissatisfied**.
You would enter **55%** for Law in the scale below.

Note: This question comes from the National Survey of College Graduates (NSCG), run by the National Science Foundation and the U.S. Census Bureau. In that survey, people were asked how satisfied they were with their job's "Contribution to Society," using options such as "Very satisfied" and "Somewhat satisfied." For this question, we group those who said "Very" or "Somewhat" satisfied into one category ("Satisfied") and the others into another ("Dissatisfied").

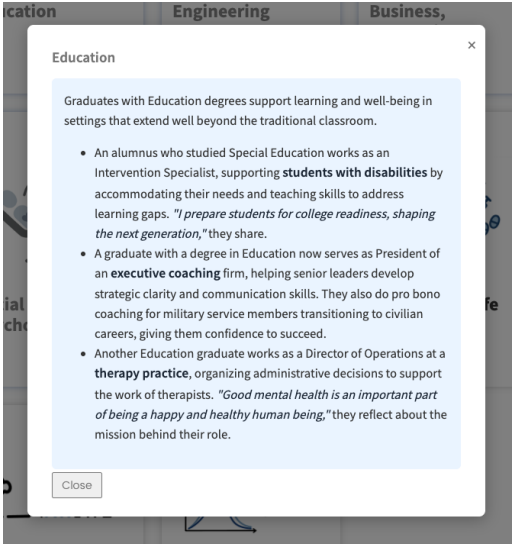
back (design) back (results)

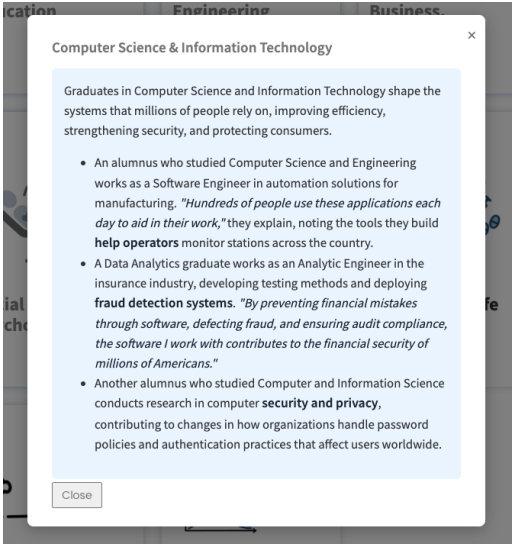
Treatment - Stories



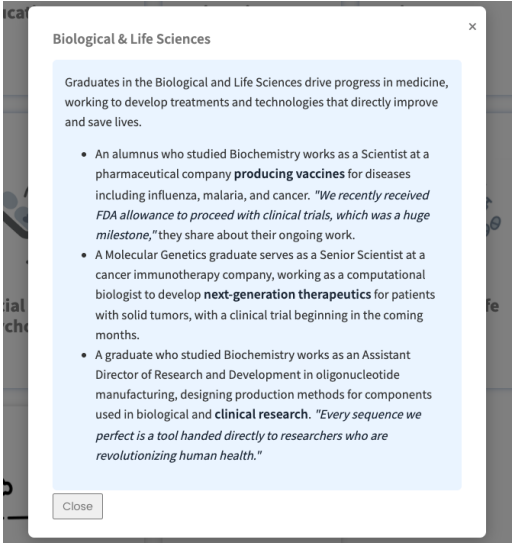


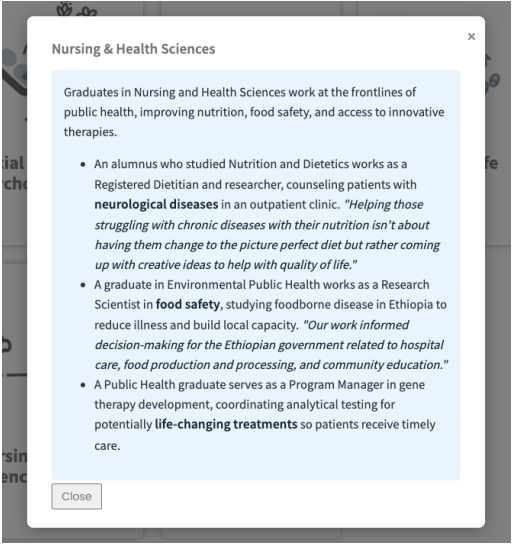
Treatment - Stories



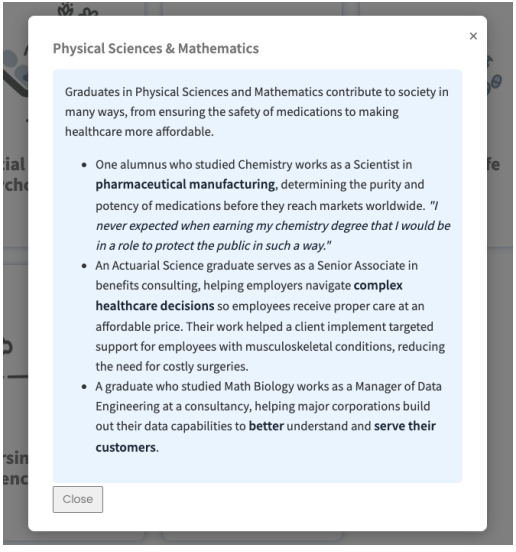


Treatment - Stories



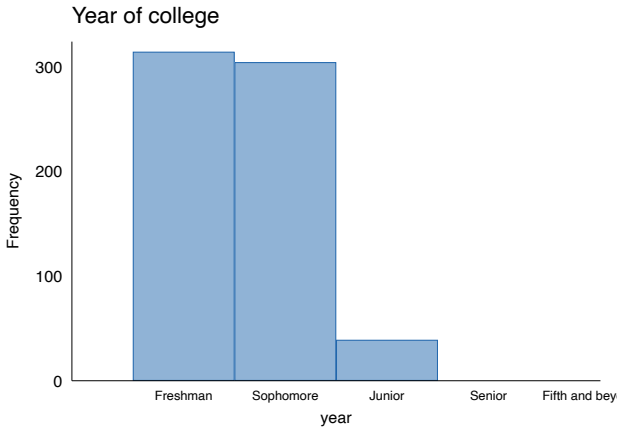


Treatment - Stories

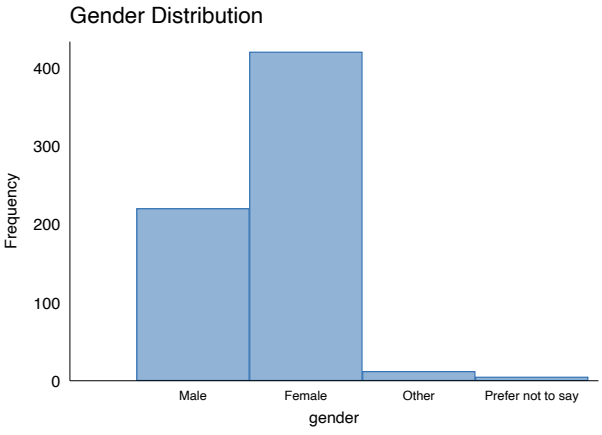


Back

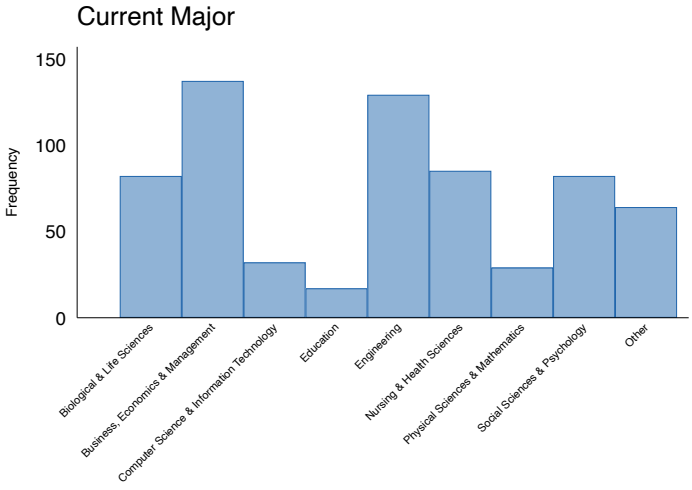
Demographics



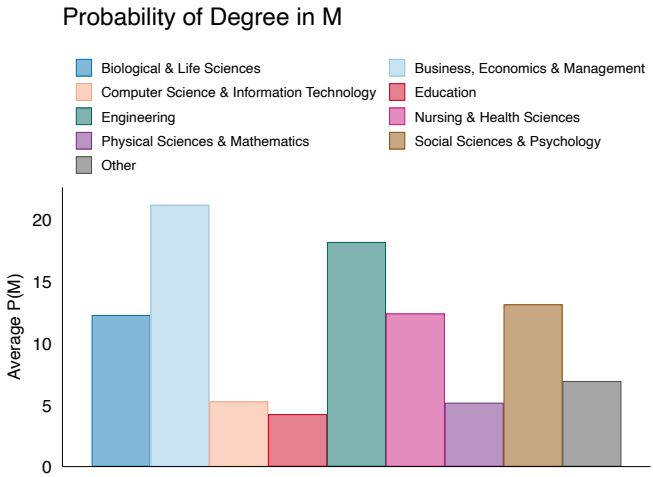
Demographics



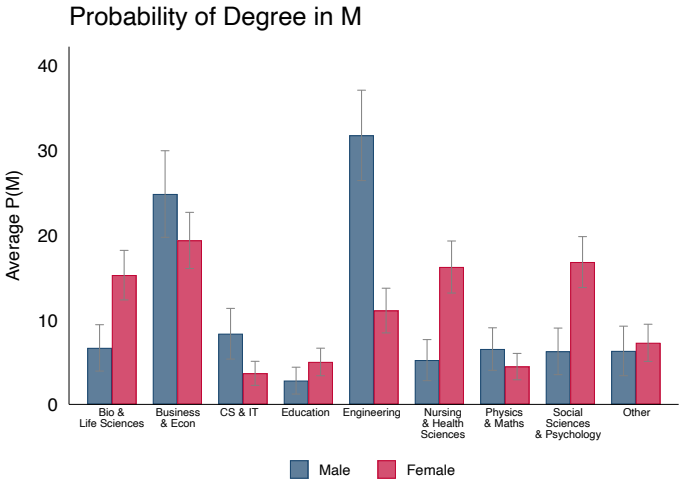
Major Intentions



Major Intentions



Major Intentions

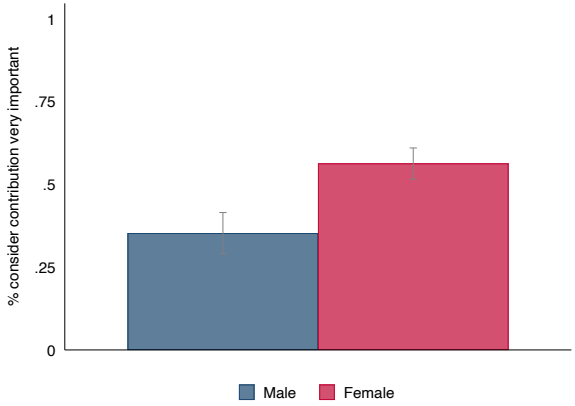


Back

Preferences

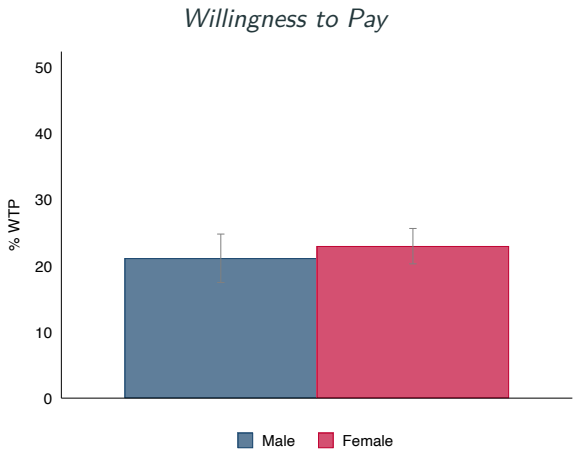
When thinking about a job, how important is each of the following factors to you?

"Contribution to Society"



Back

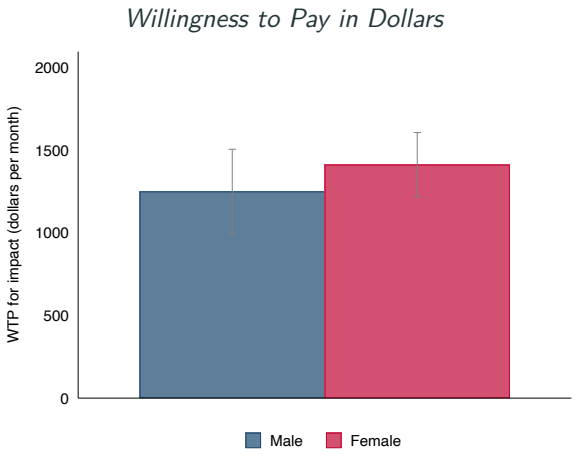
Preferences



"Now imagine that the job you are considering makes you feel satisfied with the contribution it makes to society. The job is otherwise the same as your previous one but gives you a sense of purpose through its impact. How much less pay would you be willing to accept in order to take this job? Please use the slider from 0 to 100."

Back

Preferences



"Now imagine that the job you are considering makes you feel satisfied with the contribution it makes to society. The job is otherwise the same as your previous one but gives you a sense of purpose through its impact. How much less pay would you be willing to accept in order to take this job? Please use the slider from 0 to 100."

Back

Beliefs - Self vs Population

Table 2: Correlations between Beliefs

	(1) Self Beliefs	(2) Self Beliefs	(3) Self Beliefs	(4) Self Beliefs
Population Beliefs	0.888***	0.910***		
	(0.020)	(0.024)		
Population Beliefs*Interested		−0.081**		
		(0.034)		
Interested		11.802***		7.023***
		(2.582)		(2.182)
Average Individual Beliefs			0.763***	0.769***
			(0.018)	(0.020)
AI Beliefs*Interested				−0.037
				(0.028)
Constant	10.368	4.579	20.037***	16.532**
	(9.657)	(10.113)	(6.660)	(6.867)
FEs	*Yes*	*Yes*	*Yes*	*Yes*
Observations	5256	5256	5256	5256
Adjusted R^2	0.430	0.443	0.538	0.545

Back

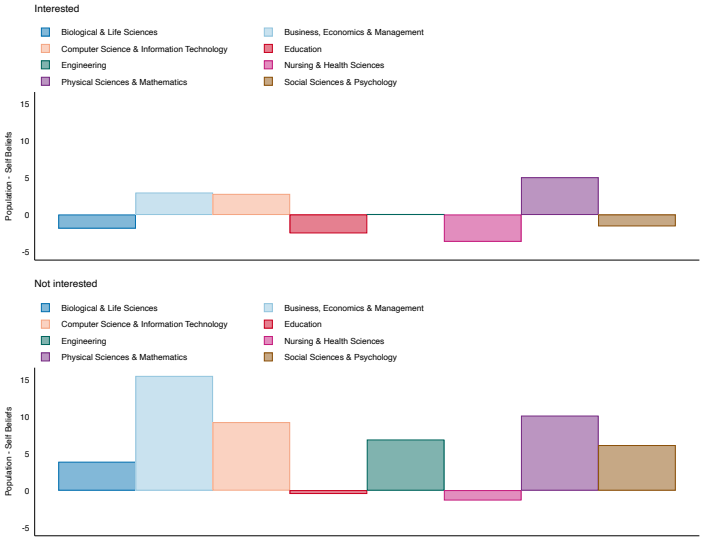
Beliefs - Self vs Population

Table 3: Self-Population Gap and Preferences for SI

	(1) Self-Pop	(2) Self-Pop	(3) Self-Pop	(4) Self-Pop
Contribution Importance	−0.830	−0.830		
	(1.345)	(1.345)		
Category Undergap (avg)		−0.332***		−0.513***
		(0.121)		(0.068)
Contr. Importance*Category Undergap		−0.352***		
		(0.133)		
Very important			0.539	0.539
			(0.888)	(0.888)
Very imp.*Category Undergap				−0.244**
				(0.102)
Constant	2.736	2.736	2.077	2.077
	(10.121)	(10.123)	(9.867)	(9.868)
FEs	*Yes*	*Yes*	*Yes*	*Yes*
Observations	5256	5256	5256	5256
Adjusted R^2	0.004	0.038	0.004	0.038

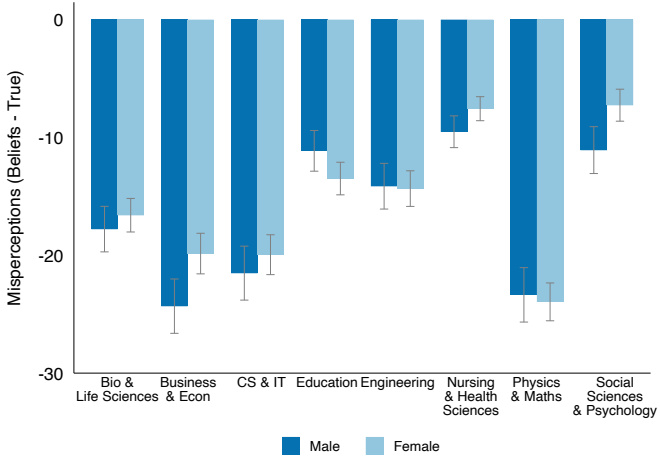
Back

Beliefs - Self vs Population



Back

Misperceptions - Gender



Back

Stereotypes - Exaggeration

Table 4: Exaggeration of Differences

	(1)
	Believed Difference
True Difference	1.576***
	(0.039)
Constant	−0.000***
	(0.000)
p(slope=1)	0.000
Observations	5256
R^2	0.350

Back

Stereotypes - Representativeness

Table 5: Population Beliefs and Representativeness

	(1) Population Beliefs
True Share	0.490***
	(0.093)
Repr. Very Dissatisfied	−10.847***
	(1.012)
Constant	39.079***
	(8.961)
Observations	5256
R^2	0.249

Back

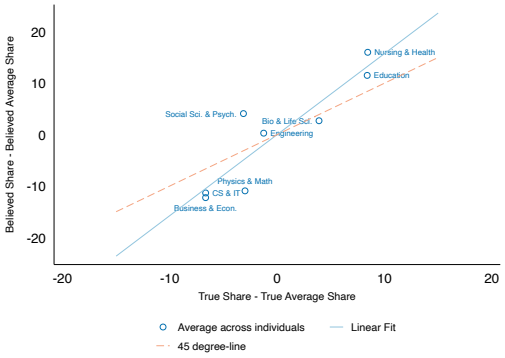
Beliefs - Salary



Back

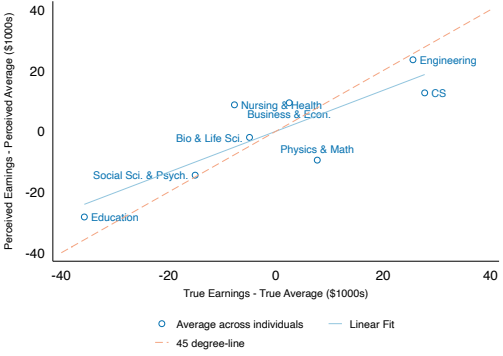
Stereotyping Social Impact vs Salary

Satisfaction with Contribution



$$\text{slope} = 1.58^{***} > 1$$

Annual Earnings



$$\text{slope} = 0.60^{***} < 1$$

Back

Choice Model Estimates

Table 6: Impact preference $\bar{\theta}$ by gender (individual-θ model, student fixed effects).

	Winsorization of probabilities		
	[0.5, 99.5]	[1, 99]	[2, 98]
$\bar{\theta}$ Women	3.411	2.915	2.422
	(0.254)	(0.217)	(0.181)
$\bar{\theta}$ Men	2.045	1.787	1.527
	(0.285)	(0.247)	(0.209)
Difference (Women − Men)	1.365***	1.127***	0.895***
	(0.316)	(0.272)	(0.229)
Students	605	605	605

Choice Model Estimates

Table 7: Structural impact preference θ_i by gender and stated contribution importance.

	[0.5, 99.5]			[1, 99]			[2, 98]		
Contribution to society is...	W	M	W−M	W	M	W−M	W	M	W−M
Not important	-1.03	1.38	-2.41	-0.87	1.35	-2.22	-0.70	1.03	-1.74
Somewhat unimportant	1.97	1.26	+0.71	1.74	1.12	+0.61	1.45	0.97	+0.48
Somewhat important	2.81	1.87	+0.94	2.41	1.64	+0.77	2.02	1.42	+0.60
Very important	4.06	2.77	+1.29	3.46	2.39	+1.07	2.86	2.03	+0.83

Back

Counterfactual: Beliefs Set to Population Values

Table 8: Belief-correction counterfactual (national values): shift in major-choice probability, versus the actual treatment effect.

	Women		Men	
	Nat.	Actual	Nat.	Actual
Bio & Life Sci.	-0.11	+1.01	+0.25	+1.29
Business & Econ.	+2.28	+0.13	+1.22	-0.44
Computer Science	+1.35	+0.55	+0.65	+0.88
Education	+0.07	+1.60	-0.14	+0.44
Engineering	-0.29	-0.73	-2.00	-0.05
Health	-1.74	+0.74	-0.51	+1.47
Physics & Maths	+1.97	+0.22	+1.09	+0.65
Social Sci. & Psych.	-1.92	-1.68	-0.27	-4.25
Other	-1.63	-0.71	-0.29	+1.14
Corr. with actual	0.37		0.09	

Back

Predictions

A shift in preferences moves students toward majors with a greater prior probability of contribution: the *rank* of revisions in choice follows the *rank* of priors in beliefs.

Predictions

A shift in preferences moves students toward majors with a greater prior probability of contribution: the *rank* of revisions in choice follows the *rank* of priors in beliefs.

Raising the impact weight by one unit ($\Delta\theta = 1$, beliefs fixed) reallocates choices (pp, women):

Away $\downarrow$		**Toward** $\uparrow$	
Business & Econ.	-0.7	Health	$+0.6$
Physics & Maths	-0.3	Social Sciences	$+0.4$
Computer Science	-0.3	Education	$+0.3$

Back

Preference Counterfactual vs. Control Change

Table 9: Preference counterfactual: shift in major-choice probability from a one-unit rise in the impact preference, versus the actual change in the control group.

	Women		Men	
	$\Delta\theta = 1$	Control	$\Delta\theta = 1$	Control
Bio & Life Sci.	+0.07	-1.50	+0.05	-0.83
Business & Econ.	-0.71	-0.36	-0.72	-0.11
Computer Science	-0.30	-0.40	-0.31	-0.66
Education	+0.25	-0.33	+0.34	+0.16
Engineering	+0.12	-0.10	+0.62	-0.41
Health	+0.58	+0.36	+0.41	+1.33
Physics & Maths	-0.31	-1.10	-0.35	-0.79
Social Sci. & Psych.	+0.43	+1.24	+0.08	+2.91
Other	-0.12	+1.09	-0.13	-1.68
Corr. (CF, control)	0.39		0.29	
Students	166		75	

Back

Control Movement in Preferences

Table 10: Impact preference $\bar{\theta}$ before and after the survey: control group.

	Pre	Post	Δ
All	2.69	3.55	0.87***
	(0.31)	(0.36)	(0.32)
Women	3.23	4.20	0.98**
	(0.36)	(0.43)	(0.39)
Men	1.67	2.33	0.66*
	(0.43)	(0.51)	(0.36)

Back

Beliefs and Initial Intentions

Table 11: Predicting Prior Major Intentions

	(1) P(M)	(2) P(M)	(3) P(M)	(4) P(M)
Prior Self Beliefs	0.182***		0.015	0.004
	(0.013)		(0.040)	(0.039)
Population Beliefs		0.061***		
		(0.018)		
Contribution Importance*Self Beliefs			0.197***	0.188***
			(0.042)	(0.041)
Contribution Importance			−12.529***	−12.911***
			(2.683)	(2.564)
Self Beliefs Log(Salary)				8.825***
				(0.966)
Expected GPA				6.023***
				(0.434)
Constant	−0.412	7.329***	10.273***	−109.492***
	(0.853)	(1.267)	(2.573)	(11.441)
FEs	*Yes*	*Yes*	*Yes*	*Yes*
Observations	5256	5256	5256	5022
Adjusted R^2	0.026	0.000	0.029	0.069

Back

Treatment Effects - Balance

	Control	Treatment	Diff.
1st Year	0.446	0.506	-0.059
2nd Year	0.482	0.446	0.036
3rd Year	0.072	0.049	0.023
Columbus	0.899	0.909	-0.010
Lima	0.007	0.011	-0.005
Mansfield	0.013	0.017	-0.004
Marion	0.020	0.011	0.008
Newark	0.059	0.046	0.013
Wooster	0.003	0.006	-0.002
Female	0.651	0.657	-0.006
Age	19.277	19.071	0.205**
Unsure	0.733	0.697	0.036

	Control	Treatment	Diff.
Contribution Important	0.857	0.854	0.002
P(Biology & Life Sciences)	11.770	12.708	-0.938
P(Business & Economics)	23.565	19.151	4.414
P(Computer Science)	5.515	5.067	0.448
P(Education)	5.090	3.497	1.593
P(Engineering)	14.129	21.759	-7.630***
P(Health Sciences)	12.498	12.314	0.184
P(Physics & Maths)	5.311	5.047	0.264
P(Social Sciences)	13.282	13.011	0.271
P(Other)	7.349	6.557	0.791

* $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Back

Attrition: Attentive vs. Inattentive

Attentive (N=657) vs Inattentive (N=320)

	Inattentive	Attentive	Diff.
Treatment	0.450	0.533	-0.083**
1st Year	0.519	0.478	0.041
2nd Year	0.450	0.463	-0.013
3rd Year	0.031	0.059	-0.028**
Columbus	0.875	0.904	-0.029
Lima	0.009	0.009	0.000
Mansfield	0.028	0.015	0.013
Marion	0.022	0.015	0.007
Newark	0.050	0.052	-0.002
Wooster	0.016	0.005	0.011
Female	0.603	0.654	-0.051
Age	19.141	19.167	-0.027
Contribution Important	0.856	0.855	0.001

	Inattentive	Attentive	Diff.
P(Biology & Life Sciences)	11.711	12.270	-0.559
P(Business & Economics)	23.947	21.214	2.733
P(Computer Science)	5.244	5.276	-0.033
P(Education)	4.303	4.241	0.062
P(Engineering)	15.606	18.193	-2.587
P(Health Sciences)	15.488	12.400	3.087
P(Physics & Maths)	2.211	5.170	-2.960***
P(Social Sciences)	12.384	13.137	-0.753
P(Other)	7.906	6.927	0.979

* p<0.10, ** p<0.05, *** p<0.01. Diff. = Inattentive − Attentive.

Attrition: Attentive vs. Inattentive (Robustness Sample)

Attentive (N=708) vs Inattentive (N=269)

	Inattentive	Attentive	Diff.
Treatment	0.506	0.506	-0.000
1st Year	0.502	0.487	0.015
2nd Year	0.465	0.456	0.008
3rd Year	0.033	0.056	-0.023
Columbus	0.881	0.900	-0.019
Lima	0.007	0.010	-0.002
Mansfield	0.026	0.017	0.009
Marion	0.022	0.016	0.007
Newark	0.048	0.052	-0.004
Wooster	0.015	0.006	0.009
Female	0.591	0.655	-0.064*
Age	19.156	19.160	-0.003
Contribution Important	0.851	0.857	-0.006

	Inattentive	Attentive	Diff.
P(Biology & Life Sciences)	11.578	12.280	-0.702
P(Business & Economics)	23.252	21.674	1.578
P(Computer Science)	4.844	5.426	-0.582
P(Education)	4.729	4.084	0.645
P(Engineering)	16.855	17.532	-0.677
P(Health Sciences)	14.706	12.919	1.787
P(Physics & Maths)	2.426	4.876	-2.450***
P(Social Sciences)	13.041	12.834	0.207
P(Other)	7.309	7.225	0.084

* p<0.10, ** p<0.05, *** p<0.01. Diff. = Inattentive − Attentive.

Back

Information Shrinks Perceived Differences across Fields

Table 12: Treatment Effects on Self Beliefs

	(1) Post SB	(2) Post SB
Prior Self Beliefs	0.617***	0.535***
	(0.016)	(0.018)
Treatment	12.563***	6.255***
	(0.706)	(0.773)
Treatment*Underestimation gap		0.384***
		(0.038)
Underestimation gap		−0.434***
		(0.028)
Constant	21.340***	33.554***
	(6.655)	(6.167)
FEs	*Yes*	*Yes*
Control Mean	66.4	66.4
Observations	5256	5256
Adjusted R^2	0.586	0.626

Back

Treatment Effects - Beliefs

Table 13: Revision of Self Beliefs by Social Desirability Bias

	(1) Low SDB	(2) High SDB	(3) Interaction
Treatment	6.507***	5.878***	7.755***
	(1.046)	(1.163)	(1.529)
Treatment*Underestimation gap	0.368***	0.392***	0.294***
	(0.051)	(0.056)	(0.082)
Treatment*SDB			−0.642
			(0.575)
Treatment*Undergap*SDB			0.040
			(0.031)
Prior Self Beliefs	0.541***	0.527***	0.535***
	(0.021)	(0.031)	(0.018)
Constant	25.705**	35.402***	31.969***
	(10.547)	(9.070)	(6.269)
FEs	*Yes*	*Yes*	*Yes*
Observations	2976	2280	5256
Adjusted R^2	0.617	0.640	0.626

Back

Revision of Beliefs by Self vs Population Beliefs

Table 14: Revision of Beliefs by Self vs Population Beliefs

	(1) Post SB	(2) Post SB
Prior Self Beliefs	0.596***	0.597***
	(0.024)	(0.025)
Treatment*Self≥Pop	9.110***	2.358***
	(0.723)	(0.773)
Treatment*Self<Pop	15.723***	9.770***
	(0.946)	(1.001)
Treatment*Underestimation gap		0.398***
		(0.038)
Constant	26.146***	29.707***
	(6.405)	(6.351)
Group FEs	*Yes*	*Yes*
FEs	*Yes*	*Yes*
Observations	5256	5256
Adjusted R^2	0.613	0.636

Back

Advice to Peers

We are recruiting 10 first-year undergraduate **students who have not yet chosen their major**. Imagine you are giving advice to one of them — a student who cares about **contribution to society** as an important part of their future career.

You'll be shown a list of majors. For each one, please rate **how strongly you would recommend it to this student who cares about contribution to society.**

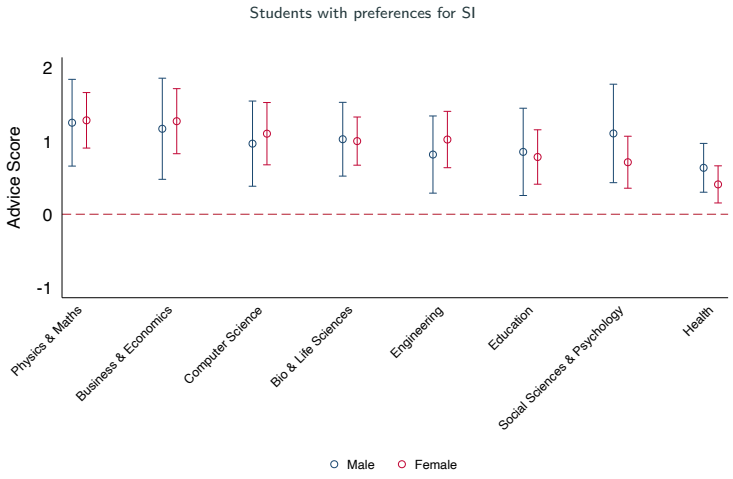
👉 **Important:**
There is a **real chance** that your advice will be randomly selected and actually shared with one of these 10 students. That means your ratings could influence someone's real decision — please answer carefully.

Using a scale from **0 to 10**, rate how much you would recommend each of the following majors to a student who cares about contribution to society:

- **0 = Not at all recommended**
- **10 = Highly recommended**

back (design) back (results)

Treatment Effects - Advice



Back

Treatment Effects - Advice

Table 15: Treatment Effects on Advice Score - Preferences for SI

	(1) Scale	(2) Scale
Treatment	0.964***	0.409***
	(0.085)	(0.103)
Underestimation gap		−0.084***
		(0.004)
Treatment*Underestimation gap		0.031***
		(0.006)
Constant	7.370***	8.279***
	(0.754)	(0.711)
Control Mean	6.6	6.6
FEs	*Yes*	*Yes*
Observations	4496	4496
Adjusted R^2	0.046	0.284

Back

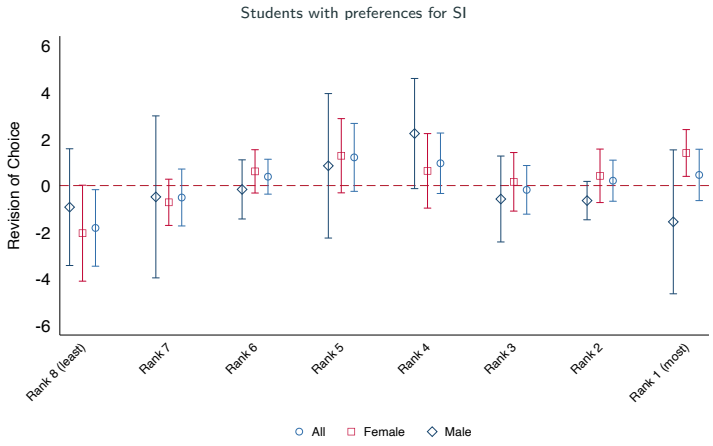
Treatment Effects - Advice

Table 16: Advice by Social Desirability Bias - Preferences for SI

	(1) Low SDB	(2) High SDB	(3) Interaction
Treatment	0.526***	0.308**	0.537**
	(0.149)	(0.147)	(0.230)
Treatment*Underestimation gap	0.022***	0.039***	0.014
	(0.008)	(0.007)	(0.013)
Treatment*SDB			−0.051
			(0.080)
Treatment*Undergap*SDB			0.007
			(0.004)
Constant	8.772***	7.064***	8.072***
	(1.154)	(0.809)	(0.747)
Control Mean	6.6	6.6	6.6
FEs	*Yes*	*Yes*	*Yes*
Observations	2464	2032	4496
Adjusted R^2	0.238	0.345	0.285

Back

Treatment Effects - Choice



Treatment Effects - Choice

Table 17: Misperception Rank and Treatment - Preferences for SI

	Final P(M)		
	(1)	(2)	(3)
	All	Women	Men
Initial P(M)	0.950***	0.956***	0.937***
	(0.012)	(0.012)	(0.025)
Treatment	0.825**	1.315***	-0.184
	(0.387)	(0.470)	(0.634)
Misperception Rank (0=highest)	0.306***	0.383***	0.143
	(0.084)	(0.110)	(0.121)
Treatment * Misp. Rank	-0.208*	-0.313**	0.016
	(0.107)	(0.123)	(0.199)
Prior credits in cat. (AU25+SP26)	0.120**	0.075	0.214**
	(0.049)	(0.056)	(0.100)
Constant	-0.958	-1.898*	2.592
	(0.822)	(1.052)	(2.022)
FEs	Yes	Yes	Yes
Observations	4496	3104	1392
Adjusted R^2	0.937	0.944	0.922

Back

Treatment Effects - Choice

Table 18: Normalized Misperception Rank — Interested Majors (Preferences for SI)

	Controlling for k		Without k	
	(1) Women	(2) Men	(3) Women	(4) Men
Initial P(M)	0.969***	0.938***	0.969***	0.938***
	(0.014)	(0.037)	(0.014)	(0.034)
Treatment	3.332**	-1.821	2.146**	-1.053
	(1.567)	(1.394)	(1.066)	(1.238)
Normalized rank (0=most, 1=least underest.)	4.417***	-0.090	4.634***	-0.207
	(1.504)	(0.865)	(1.508)	(0.878)
Treatment*Norm. rank	-3.152*	0.533	-3.543*	0.796
	(1.865)	(3.026)	(1.832)	(2.823)
k (# interested majors)	0.176	-0.094		
	(0.182)	(0.157)		
Treatment*k	-0.322	0.218		
	(0.230)	(0.287)		
Constant	-3.710	-6.288	-3.277	-6.524
	(4.147)	(10.004)	(3.885)	(9.895)
FEs	Yes	Yes	Yes	Yes
Observations	1017	465	1017	465
Adjusted R^2	0.933	0.923	0.933	0.923

Back

New vs Already Considered Majors

Table 19: Treatment Effect by Misperception and Prior Interest - Women, Preferences for SI

	Final P(M)	
	(1)	(2)
	Already considered	New
Initial P(M)	0.958***	
	(0.015)	
Treatment	0.795	−0.046
	(0.582)	(0.183)
Demeaned Undergap	−0.142***	−0.040
	(0.039)	(0.026)
Demeaned Undergap*Treatment	0.094**	0.039
	(0.042)	(0.026)
Prior credits in cat. (AU25+SP26)	0.089	0.056
	(0.079)	(0.052)
Constant	−0.894	−0.317
	(3.665)	(0.882)
FEs	*Yes*	*Yes*
Observations	1017	2087
Adjusted R^2	0.933	0.008

Back

New Interest: From Zero to Positive Probability

Table 20: New Interest by Misperception - Preferences for SI

	P(M) from 0 to positive		
	(1)	(2)	(3)
	All	Women	Men
Treatment	−0.0001	−0.0038	0.0030
	(0.0093)	(0.0114)	(0.0169)
Demeaned Undergap	−0.0015***	−0.0016***	−0.0014*
	(0.0004)	(0.0006)	(0.0008)
Demeaned Undergap*Treatment	0.0013**	0.0018**	0.0004
	(0.0006)	(0.0008)	(0.0010)
Prior credits in cat. (AU25+SP26)	0.0014	0.0012	0.0016
	(0.0013)	(0.0014)	(0.0028)
Constant	0.0755	0.0820	0.0920
	(0.0868)	(0.0934)	(0.2591)
Control mean	0.044	0.043	0.044
FEs	*Yes*	*Yes*	*Yes*
Observations	3014	2087	927
Adjusted R^2	0.009	0.016	−0.001

Back

Probability Mass by Self vs Population Beliefs

Table 21: Probability Mass by Self vs Population Beliefs - Women, Preferences for SI

	Self≥Pop		Self<Pop	
	(1)	(2)	(3)	(4)
	Above mean	Below mean	Above mean	Below mean
Treatment	0.691	0.046	1.621**	−0.713
	(0.763)	(1.361)	(0.810)	(0.775)
Control mean prior mass	26.44	38.99	8.83	17.95
Control mean post mass	25.40	39.53	7.25	18.30
FEs	*Yes*	*Yes*	*Yes*	*Yes*
Observations	388	388	388	388

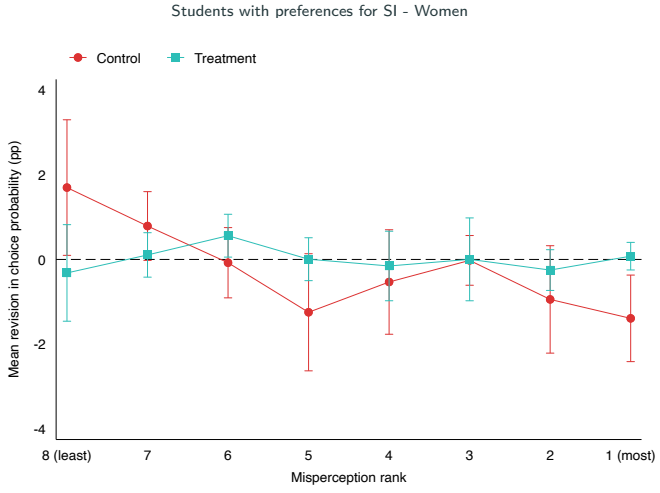
Back

Treated Women Are More Likely to Choose More Misperceived Fields

Students with preferences for SI

	Final P(M)		
	(1)	(2)	(3)
	All	Women	Men
Initial P(M)	0.949***	0.955***	0.936***
	(0.012)	(0.013)	(0.026)
Treatment	0.098	0.217	-0.127
	(0.115)	(0.151)	(0.155)
Demeaned Undergap	-0.066***	-0.074***	-0.045
	(0.020)	(0.025)	(0.035)
Demeaned Undergap*Treatment	0.042**	0.060**	0.003
	(0.021)	(0.024)	(0.041)
Prior credits in cat. (AU25+SP26)	0.124**	0.079	0.219**
	(0.050)	(0.057)	(0.101)
Constant	0.109	-0.562	3.077
	(0.764)	(0.886)	(2.324)
FEs	Yes	Yes	Yes
Observations	4496	3104	1392
Adjusted R^2	0.937	0.944	0.922

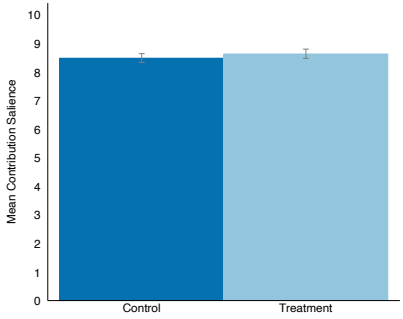
Control Women Move Away from Misperceived Majors



Salience Preferences Back

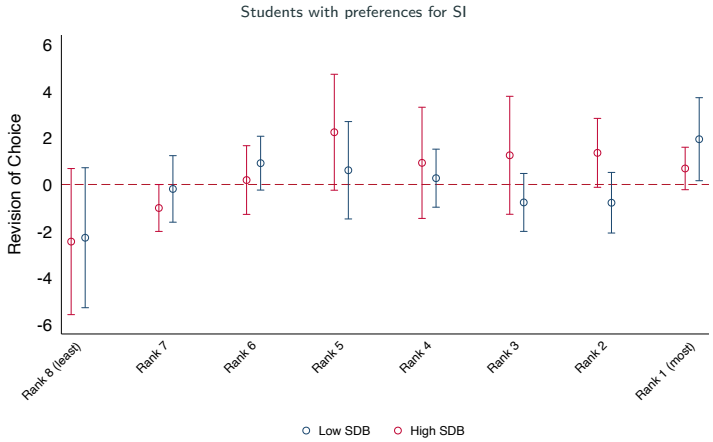
Salience

On a scale from 0 to 10, how much was the idea of contributing to society on your mind during the survey?



Back

Treatment Effects - Women



Back

Treatment Effects - Major Intentions

Table 22: Treatment Effect by Misperception and Self-Reported GPA - Preferences for SI

	Median split			Continuous		
	(1) All	(2) Women	(3) Men	(4) All	(5) Women	(6) Men
Initial P(M)	0.949***	0.954***	0.936***	0.949***	0.955***	0.936***
	(0.012)	(0.013)	(0.025)	(0.012)	(0.013)	(0.025)
Demeaned Undergap*Treatment	0.056*	0.069*	0.004	0.040**	0.053***	−0.001
	(0.033)	(0.040)	(0.030)	(0.020)	(0.019)	(0.037)
Demeaned Undergap*Treatment*Above-median GPA	−0.027	−0.019	−0.014			
	(0.043)	(0.044)	(0.073)			
Demeaned Undergap*Treatment*GPA (std.)				−0.041	−0.049	0.014
				(0.041)	(0.047)	(0.036)
Prior credits in cat. (AU25+SP26)	0.125**	0.082	0.218**	0.124**	0.079	0.217**
	(0.050)	(0.058)	(0.100)	(0.049)	(0.056)	(0.100)
FEs	*Yes*	*Yes*	*Yes*	*Yes*	*Yes*	*Yes*
Observations	4488	3096	1392	4488	3096	1392
Adjusted R^2	0.937	0.944	0.922	0.937	0.944	0.922

Back

Treatment Effects - Major Intentions

Table 23: Probability Mass by Relative Expected GPA - Women, Preferences for SI

	Relatively good GPA		Relatively bad GPA	
	(1)	(2)	(3)	(4)
	Above mean	Below mean	Above mean	Below mean
Treatment	1.907*	0.295	0.387	−0.434
	(1.005)	(1.385)	(0.452)	(0.771)
Control mean prior mass	24.11	46.78	11.06	10.10
Control mean post mass	22.06	46.84	10.56	10.59
FEs	*Yes*	*Yes*	*Yes*	*Yes*
Observations	375	375	375	375

Back

Treatment Effects - Salary

Table 24: Revision Salary - Preferences for SI

	(1) Post Earnings	(2) Post Earnings
Self Beliefs Log(Salary)	0.616***	0.614***
	(0.048)	(0.048)
Treatment	0.061***	0.060**
	(0.017)	(0.024)
Treatment*Underestimation gap		−0.000
		(0.001)
Underestimation gap		−0.001**
		(0.001)
Constant	4.093***	4.129***
	(0.568)	(0.567)
FEs	*Yes*	*Yes*
Observations	4496	4496
Adjusted R^2	0.404	0.406

Back

Treatment Effects - WTP

Table 25: Willingness to Pay - Preferences for SI

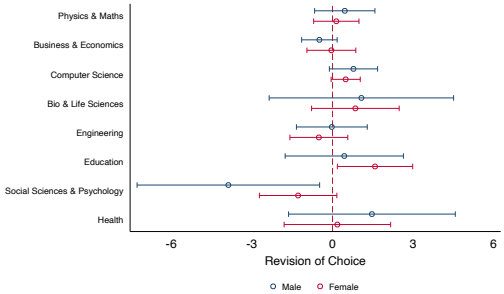
	(1) WTP	(2) WTP
Treatment	−1.092	−1.118
	(2.536)	(2.537)
Post	−0.445	−0.450
	(2.608)	(2.609)
Post*Treatment	1.959	1.967
	(3.555)	(3.556)
Contribution Salience		0.137
		(0.663)
Constant	17.465	16.708
	(19.105)	(19.537)
Control Mean	24.2	24.2
FEs	*Yes*	*Yes*
Observations	562	562
Adjusted R^2	−0.008	−0.009

Back

Intentions Maps into Real Choices

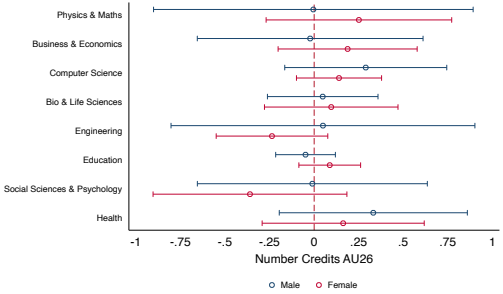
Survey

Treatment effect on reported intentions $P(\text{major})$



Administrative data

Treatment effect on # AU26 credits, by field

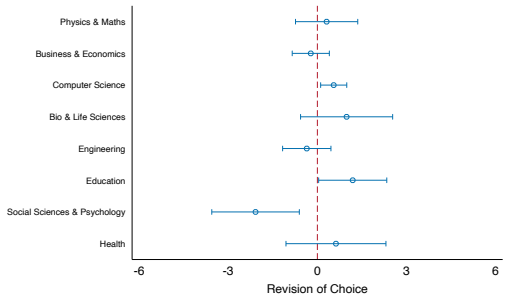


Back

Intentions Maps into Real Choices - First Snapshot

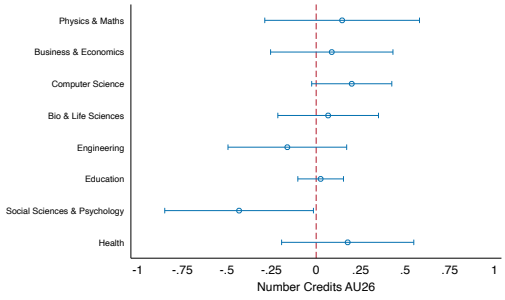
Survey

Treatment effect on reported intentions $P(\text{major})$



Administrative data

Treatment effect on # AU26 credits, by field

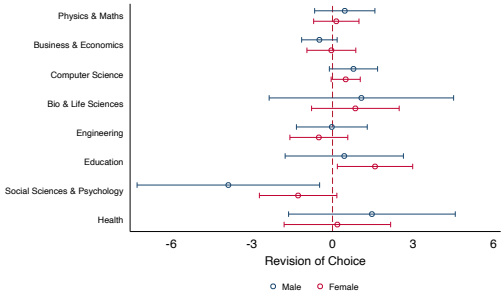


Back

Intentions Maps into Real Choices - First Snapshot

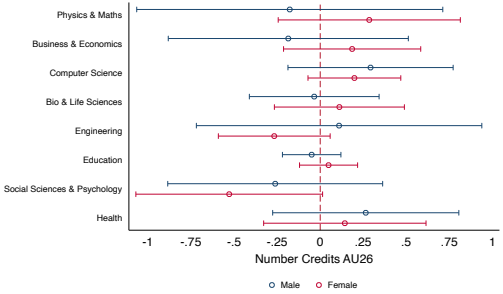
Survey

Treatment effect on reported intentions $P(\text{major})$



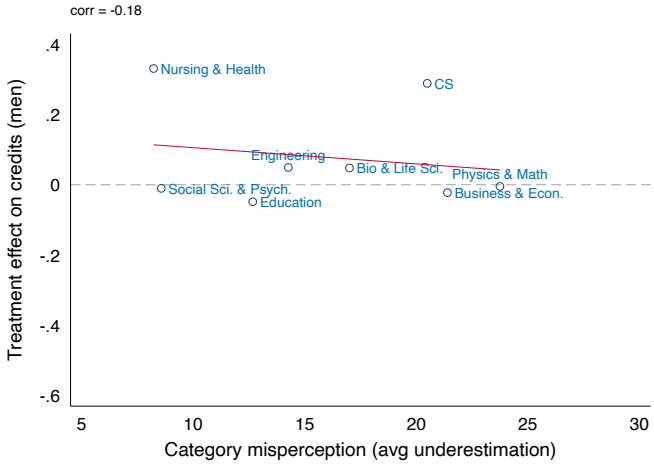
Administrative data

Treatment effect on # AU26 credits, by field



Back

Treated Women Enroll in the Most Systematically Underestimated Fields



Back

Table 26: Admin Credits (AU26) in the Most Underestimated Fields - Preferences for SI

	(1) All	(2) Women	(3) Men
Treatment	0.428	0.625*	−0.080
	(0.277)	(0.324)	(0.563)
Prior credits in 3 fields (AU25+SP26)	0.235***	0.246***	0.210***
	(0.027)	(0.031)	(0.058)
Initial P(3 fields)	0.050***	0.048***	0.056***
	(0.005)	(0.006)	(0.010)
Constant	−2.213	−3.023	−1.136
	(2.738)	(2.886)	(9.861)
Control Mean	5.13	4.59	6.30
FEs	*Yes*	*Yes*	*Yes*
Observations	548	376	172
Adjusted R^2	0.511	0.505	0.486

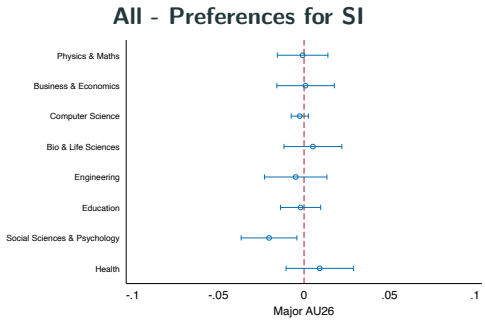
Back

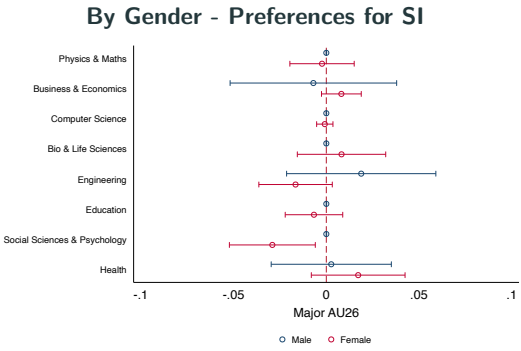
Classes and Reported Intentions

Table 27: Credits (AU26) - Preferences for SI

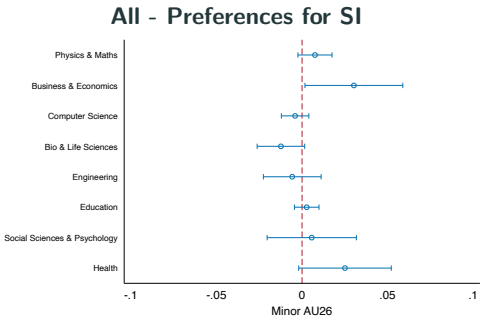
	Baseline			+ Prior intention		
	(1)	(2)	(3)	(4)	(5)	(6)
	All	Women	Men	All	Women	Men
Prior credits in cat. (AU25+SP26)	0.297***	0.307***	0.271***	0.298***	0.306***	0.275***
	(0.011)	(0.013)	(0.022)	(0.011)	(0.014)	(0.022)
Final P(M)	0.040***	0.036***	0.049***	0.041***	0.030***	0.061***
	(0.003)	(0.003)	(0.005)	(0.008)	(0.010)	(0.013)
Initial P(M)				−0.002	0.006	−0.013
				(0.008)	(0.010)	(0.012)
Constant	−0.221	−0.258	−0.036	−0.222	−0.258	−0.045
	(0.275)	(0.306)	(0.668)	(0.275)	(0.305)	(0.668)
FEs	*Yes*	*Yes*	*Yes*	*Yes*	*Yes*	*Yes*
Observations	4932	3384	1548	4932	3384	1548
Adjusted R^2	0.573	0.553	0.617	0.573	0.553	0.618

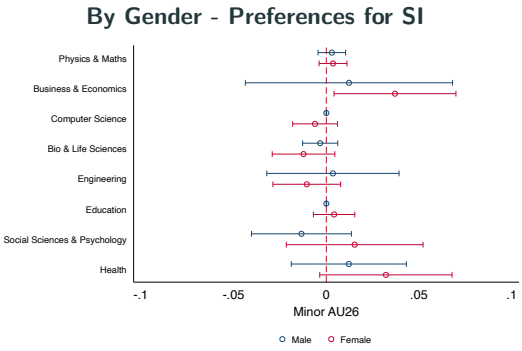
Major Declarations





Minor Declarations





Back

More on Beliefs

- We define π^i_m as the *actual* probability of being satisfied with contribution after graduating in m:

$$\pi^i_m = Pr(c = 1|m)$$

- Given c_m, the contribution level in field m, individual i is happy with their contribution if it is higher than a certain threshold:

$$\pi^i_m = Pr(c = 1|m) = Pr(c_m > \bar{c}^i)$$

- The threshold is individual specific and related to preferences:

$$\bar{c}^i = f(\theta^i) \Rightarrow \frac{\partial \bar{c}^i}{\partial \theta^i} > 0 \quad \wedge \quad \frac{\partial^2 \bar{c}^i}{\partial (\theta^i)^2} < 0$$

More on Beliefs

- Contribution in major m is distributed as: $c_m \sim \mathcal{N}(\lambda_m, \sigma_m^2)$
- If individual i had the correct beliefs on the distribution of c_m:

$$\pi_m^i = Pr(c_m > \bar{c}_i) = \Phi\left(\frac{\lambda_m - \bar{c}^i}{\sigma_m}\right)$$

- We denote i's perceived average $\tilde{\lambda}_m^i$ and perceived variance $\tilde{\sigma}_m^{2,i}$
- The *perceived* probability of being satisfied with contribution is:

$$\tilde{\pi}_m^i = \Phi\left(\frac{\tilde{\lambda}_m^i - \bar{c}^i}{\tilde{\sigma}_m^i}\right)$$

$\Rightarrow$ *Our intervention aims at moving both* $\tilde{\lambda}_m^i$ *and* $\tilde{\sigma}_m^i$

Selection

- Probability that *graduates* in m are satisfied, with $\bar{c}^m$ their average threshold:

$$\pi_m^m = Pr(c_m > \bar{c}^m)$$

- If i believes graduates in m care less about contribution than i does, i attributes to them a lower threshold:

$$\mathbb{E}^i\left[\theta^j \mid q_m^j = 1\right] < \theta^i \quad \Rightarrow \quad \tilde{\bar{c}}^m < \bar{c}^i$$

- For a given perceived distribution $\tilde{c}_m$, i sees graduates as more likely satisfied than themselves:

$$\tilde{\pi}_m^m = Pr(\tilde{c}_m > \tilde{\bar{c}}^m) > Pr(\tilde{c}_m > \bar{c}^i) = \tilde{\pi}_m^i$$

$\Rightarrow$ *We elicit both* $\tilde{\pi}_m^m$ *and* $\tilde{\pi}_m^i$

Selection

- The intervention reveals the true π_m^m. Suppose i *underestimated* it: $\tilde{\pi}_m^m < \pi_m^m$
- The model admits two explanations:
  - **Case 1 – misperceived threshold:** graduates in m care even less than i thinks,

    $$\bar{c}^m < \tilde{\bar{c}}^m$$

    learning the truth tells i only that graduates care less $\Rightarrow$ own $\tilde{\pi}_m^i$ *unchanged*
  - **Case 2 – misperceived distribution:** $(\tilde{\lambda}_m^i, \tilde{\sigma}_m^i) \neq (\lambda_m, \sigma_m)$,
    the truth reveals contribution is better than i thought $\Rightarrow$ i revises $\tilde{\lambda}_m^i, \tilde{\sigma}_m^i$ and own $\tilde{\pi}_m^i$

$\Rightarrow$ *Only Case 2 shifts i's own beliefs and choice*

Back

Ariely, D., Kamenica, E., and Prelec, D. (2008). Man's search for meaning: The case of legos. *Journal of Economic Behavior & Organization*, 67(3-4):671–677.

Ashraf, N., Bandiera, O., Minni, V., and Zingales, L. (2024). Meaning at work. *Unpublished Working Paper, London School of Economics*.

Astorne-Figari, C. and Speer, J. D. (2018). Drop out, switch majors, or persist? the contrasting gender gaps. *Economics Letters*, 164:82–85.

Bordalo, P., Coffman, K., Gennaioli, N., and Shleifer, A. (2016). Stereotypes. *The Quarterly Journal of Economics*, 131(4):1753–1794.

Bordalo, P., Coffman, K., Gennaioli, N., and Shleifer, A. (2019). Beliefs about gender. *American Economic Review*, 109(3):739–773.

Boucher, K. L., Fuesting, M. A., Diekman, A. B., and Murphy, M. C. (2017). Can i work with and help others in this field? how communal goals influence interest and participation in stem fields. *Frontiers in psychology*, 8:239640.

Breda, T., Grenet, J., Monnet, M., and Van Effenterre, C. (2023). How effective are female role models in steering girls towards STEM? evidence from French high schools. *The Economic Journal*, 133(653):1773–1809.

Brief, A. P. and Nord, W. R., editors (1990). *Meanings of Occupational Work: A Collection of Essays*. Issues in Organization and Management Series. Lexington Books, Lexington, MA.

Burbano, V., Padilla, N., and Meier, S. (2024a). Gender differences in preferences for meaning at work. *American Economic Journal: Economic Policy*, 16(3):61–94.

Burbano, V. C., Folke, O., Meier, S., and Rickne, J. (2024b). The gender gap in meaningful work. *Management Science*, 70(10):7004–7023.

Cassar, L. and Meier, S. (2018). Nonmonetary incentives and the implications of work as a source of meaning. *Journal of Economic Perspectives*, 32(3):215–238.

Conlon, J. J. (2021). Major malfunction: A field experiment correcting undergraduates' beliefs about salaries. *Journal of Human Resources*, 56(3):922–939.

Conlon, J. J. and Patel, D. (2026). What jobs come to mind? stereotypes about fields of study. *The Quarterly Journal of Economics*, page qjag031.

Diekman, A. B., Clark, E. K., Johnston, A. M., Brown, E. R., and Steinberg, M. (2011). Malleability in communal goals and beliefs influences attraction to stem careers: evidence for a goal congruity perspective. *Journal of personality and social psychology*, 101(5):902.

Gennaioli, N. and Shleifer, A. (2010). What comes to mind. *The Quarterly journal of economics*, 125(4):1399–1433.

Kahn, S. and Ginther, D. (2018). Women and science, technology, engineering, and mathematics (stem): Are differences in education and careers due to stereotypes, interests, or family? In Averett, S. L., Argys, L. M., and Hoffman, S. D., editors, *The Oxford Handbook of Women and the Economy*, pages 767–798. Oxford University Press.

Long, M. C., Goldhaber, D., and Huntington-Klein, N. (2015). Do completed college majors respond to changes in wages? *Economics of Education Review*, 49:1–14.

Patnaik, A., Wiswall, M., and Zafar, B. (2021). College majors. In McCall, B. P., editor, *The Routledge Handbook of the Economics of Education*, pages 183–211. Routledge.

Su, R. and Rounds, J. (2015). All stem fields are not created equal: People and things interests explain gender disparities across stem fields. *Frontiers in psychology*, 6:189.

Weinstein, R. (2022). Local labor markets and human capital investments. *Journal of Human Resources*, 57(5):1498–1525.

Wiswall, M. and Zafar, B. (2015). Determinants of college major choice: Identification using an information experiment. *The Review of Economic Studies*, 82(2):791–824.

Wiswall, M. and Zafar, B. (2018). Preference for the workplace, investment in human capital, and gender. *The Quarterly Journal of Economics*, 133(1):457–507.

Wiswall, M. and Zafar, B. (2021). Human capital investments and expectations about career and family. *Journal of Political Economy*, 129(5):1361–1424.